



**Verified Carbon
Standard**

TRAMONTANA ICS-01 – THE
DISTRIBUTION OF IMPROVED
COOKSTOVES TO INDIGENOUS
COMMUNITIES ACROSS RURAL INDIA



Carbon
— C H E C K —

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Project ID	4457
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<i>VCS Standard Version</i>	VCS Standard v4.7
Current applied methodology	VMR0006 Energy efficiency and fuel switch measures in thermal applications v1.2
<i>New applied methodology</i>	VM0050 Energy efficiency and fuel switch measures in cookstoves v1.0
Client	Tramontana Impact Limited
Prepared by	Carbon Check (India) Pvt. Ltd.
Approved by	Amit Anand, CEO
Work carried out by	Manas Hadler – Team Leader/ Technical Expert Temesgen Areaya – Trainee Assessor Nisha Jha – Trainee Assessor S. Ranganathan – Technical Reviewer

Summary:

- **Description of the Verification of the Project's Requantified Monitoring Periods**

Validation of Methodology Change:

Carbon Check (India) Private Limited, hereinafter referred to as "Carbon Check", an accredited Validation and Verification Body (VVB), was engaged by the Project Proponent, Tramontana Impact Limited, to perform the validation of the methodology change for the VCS Project Activity titled "Tramontana ICS-01 – The Distribution of Improved Cookstoves to Indigenous Communities across Rural India" (VCS ID – 4457). This validation was required due to the project's transition from the previously applied methodology VMR0006 v1.2 /B06/ to the updated VM0050, Version 1.0 /B02/. The validation focused on confirming the continued eligibility of the project activity under VM0050, assessing the appropriateness of updated baseline and project scenarios, verifying the application of emission factors and monitored parameters, and reviewing modifications to the Project Description (PD). The assessment ensured that the methodology change upheld the conservativeness, transparency, and environmental integrity required under the Verified Carbon Standard (VCS) framework.

Verification of Requantified Emission Reductions:

CC IPL conducted the verification of the requantified greenhouse gas (GHG) emission reductions and removals for the monitoring period from 21-January-2023 to 23-August-2023. The verification involved a comprehensive desk-based review of the revised Project Description /01/, Requantification report /03/ the emission reduction calculation spreadsheet, and other relevant supporting documentation provided by the Project Proponent. The VVB assessed the updated monitoring procedures and quantification methods used under VM0050 v1.0, comparing the newly calculated figures against previously verified values under VMR0006 v1.2 /B06/. This process confirmed that the requantified emission reductions were accurate, conservative, and fully compliant with the requirements of VCS Version 4.7 and ISO 14064-3:2019.

Project:

The project applies the approved VCS methodology VM0050, Version 1.0 /B02/, following its transition from VMR0006 v1.2 /B06/. The project activity involves the implementation of measures aimed at achieving verifiable GHG emission reductions. The updated quantification approach and monitoring procedures have been validated and verified to ensure methodological consistency, environmental integrity, and compliance with all applicable VCS requirements.

- **Purpose and Scope of Verification and Validation Activities**

Purpose:

The purpose of the validation and requantification verification was to assess the project's transition from the previously applied methodology VMR0006 v1.2 to the updated VM0050, Version 1.0, and to confirm the accuracy and conservativeness of the requantified greenhouse gas (GHG) emission reductions. The validation aimed to evaluate the revised PD /01/, ensuring that all updates to the project boundary, baseline

scenario, emission sources, monitored parameters, and calculation procedures are fully compliant with the VCS Standard version 4.7 /B01/ and the applied methodology VM0050 v1.0 /B02/. The verification further aimed to confirm that the emission reductions presented in the VCS Requantification Report /03/ are complete, accurate, transparent, and eligible for issuance, and that they are supported by reliable and verifiable data.

Scope:

The scope of the validation and verification activities included:

- Validation of the revised Project Description, including updates to baseline and project scenarios, emissions sources, and monitoring frameworks, in accordance with VM0050 v1.0 /B02/.
- Review and assessment of the VCS Requantification Report /03/ prepared by the Project Proponent to confirm its methodological correctness and consistency with VCS requirements /B01/.
- Verification of the requantified GHG emission reductions and removals for the monitoring period from 21-January-2023 to 23-August-2023.
- Evaluation of the application of updated parameters, such as emission factors, stove usage assumptions, and baseline conditions, under VM0050 v1.0 /B02/.
- Review of the GHG emission reduction calculation spreadsheets and associated supporting documentation for data traceability and reliability.
- Assessment of the revised monitoring systems, data collection procedures, and QA/QC measures to ensure they align with the updated monitoring plan.
- Confirmation that the reported GHG reductions are conservative, substantiated by verifiable evidence, and compliant with the VCS framework for credit issuance.

1 Monitoring Periods Selected for Requantification

The following monitoring period was selected by the project proponent for requantification under the new methodology:

- **21-January-2023 to 23-August-2023**

This period was chosen as part of the first instance of the grouped project, which began stove distribution in Andhra Pradesh. The period corresponds to the early phase of the implementation and issuance of the first set of Verified Carbon Units (VCUs) under VMR0006 v1.2 /B06/.

2 Method and Criteria Used for Validation/Verification Activities

The validation and verification activities for the requantified monitoring periods were conducted in accordance with the Verified Carbon Standard (VCS) Program Rules, Version 4.7 /B01/, and relevant Verra guidance on Requantification and Reconciliation of Verified Emission Reductions and Removals. The applied methodology and tools used to determine emission reductions include:

- **VM0050:** *Energy Efficiency and Fuel Switch Measures in Cookstoves*, Version 1.0 /B02/
- **AMS-II.G:** *Energy efficiency measures in thermal applications of non-renewable biomass*, Version 13.1 (referenced under VM0050) /B02/
- **VT0008:** *Additionality Assessment*, Version 1.0 /B03/
- **CDM TOOL30:** *Calculation of the fraction of non-renewable biomass*, Version 4.0 /B03/

The verification body (VVB) adopted a risk-based and sampling-based audit approach in accordance with the VCS Standard and relevant methodological requirements. The audit began with a comprehensive desk review of the updated Project Description, the VCS Requantification Report, monitoring data, and ex-ante and ex-post calculation spreadsheets, as well as the baseline assumptions supporting the revised methodology. The review focused on validating that the project fully applied the newly adopted methodology, VM0050 (Energy Efficiency and Fuel Switch Measures in Cookstoves, v1.0) /B02/. This included verifying updates to key monitoring parameters, data sources, and confirming appropriate application of revised baseline conditions where applicable.

During the verification of monitoring data, the VVB assessed the consistency, completeness, and accuracy of the reported parameters. Monitoring records were thoroughly cross-checked against the calculation spreadsheets to ensure the integrity of reported results. Default values and emission factors were validated to confirm their alignment with the approved methodologies and tools, including AMS-II.G v13.1 and CDM TOOL30 v4.0. The VVB also conducted a detailed verification of ex-post emission reductions for the monitoring period 21-January-2023 to 23-August-2023. This involved cross-verifying all assumptions, formulas, and calculations in the spreadsheets to ensure they were consistent with the criteria laid out in the applied methodology and associated tools.

The audit process included interviews and written correspondence with the project proponent and relevant monitoring staff. These engagements helped the VVB confirm the credibility of data sources, understand monitoring practices, and evaluate the effectiveness of procedural and quality control measures implemented during the monitoring period.

3 Number and Nature of Findings Raised During the Audit

A risk-based approach has been followed to perform this verification. During the course of verification, a total of 16 findings were raised, which include:

Type	Count	Summary of Findings
Clarification Requests (CLs)	06	Refer to Appendix 5 for further details.
Corrective Action Requests (CARs)	08	Refer to Appendix 5 for further details.
Forward Action Requests (FARs)	02	Refer to Appendix 5 for further details.

All Corrective Action Requests (CARs) and Clarification Requests (CLs) raised in previous stages have been satisfactorily addressed by the Project Proponent. The two Forward Action Requests (FARs) identified during

previous joint validation and verification have also been resolved in the course of this verification. Nevertheless, two new FARs have been raised as a result of the current verification.

4 Uncertainties Associated with the Requantification Verification

Uncertainties were reviewed in detail, particularly concerning:

- Historical baseline fuel consumption data,
- Representativeness of usage survey samples,
- Stove stacking practices (continued use of traditional stoves alongside ICS), and
- Local variations in fuel moisture content and calorific value.

These uncertainties were determined to be **appropriately addressed** under VM0050 v1.0 /B02/ through conservative assumptions and sampling designs. The VVB found no material uncertainty that would significantly alter the requantified emission estimates.

5 Summary of the Requantification Verification Conclusion

Based on the review of the updated project documentation, recalculated emission reductions, and the application of VM0050 v1.0, the VVB concludes that:

- The **updated project description** and GHG quantification approach are **valid** and consistent with the VCS Standard and applicable methodology.
- The **requantified reductions** for the monitoring period 21-January-2023 to 23-August-2023 are **credible, conservative, and verifiable**.
- All **identified issues** were resolved prior to finalization of this report.

The VVB recommends approval of the updated project documentation and issuance of VCUs based on the **requantified results**.

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1 INTRODUCTION

1.1 Objective

Carbon Check has been contracted by Tramontana Impact Limited, the Project Proponent (PP), to conduct the validation of the methodology transition and requantification verification for the project titled “Tramontana ICS-01 – The Distribution of Improved Cookstoves to Indigenous Communities across Rural India” (VCS ID – 4457) for the monitoring period from 21-January-2023 to 23-August-2023 (inclusive).

The objective of this assessment is twofold:

- **Validation:** To confirm that the updates to the Project Description /01/, including changes to the baseline scenario, emission sources, monitoring approach, and calculation methodologies, are fully aligned with the requirements of VM0050, Version 1.0 /B02/ and the VCS Standard, Version 4.7 /B01/.
- **Verification:** To independently assess the accuracy, transparency, and conservativeness of the requantified GHG emission reductions for the selected monitoring period, ensuring that data are free from material misstatements and that the recalculated emission reductions remain eligible for Verified Carbon Unit (VCU) issuance.

This validation and verification exercise ensures methodological consistency and environmental integrity, supporting Verra’s ability to issue credible and robust carbon credits under the updated methodology.

1.2 Scope and Criteria

The validation and requantification verification of this project was conducted by Carbon Check and is based on the assessment of the requantification Report /03/ for the period 21-January-2023 to 23-August-2023, the revised VCS Project Description (PD) /01/, emission reduction calculation spreadsheets /02/, supporting documentation submitted by the Project Proponent, and information gathered through interviews and correspondence. Publicly available information was also considered where relevant.

The assessment covers the validation of the project’s transition from the previously applied methodology VMR0006, Version 1.2 /B06/ to the updated methodology VM0050, Version 1.0 /B02/ and the verification of the requantified GHG emission reductions reported for the monitoring period under review. A risk-based and evidence-driven approach was employed, with a focus on areas that may materially affect the conservativeness, accuracy, or completeness of the reported results.

Sources of information include:

- The revised VCS PD version 1.8, dated 02-January-2026 /01/.
- Requantification Report version 1.2, dated 02-January-2026 /03/;
- Revised GHG emission reduction calculation spreadsheets /02/;
- Supporting documentation and clarifications provided by the Project Proponent;
- Information obtained during remote or on-site interviews with project and monitoring staff.

The assessment was conducted against the following key criteria:

- VCS Standard, Version 4.7 /B01/;
- VCS Validation and Verification Manual, Version 3.2 /B01/;
- VCS Program Guide, Version 4.4 /B01/;
- Approved methodology VM0050, Version 1.0 /B02/ and AMS-II.G v13.1 /B02/;
- VCS Methodology Change and Requantification Procedure, Version 4.0 /B01/;
- Previously applied methodology VMR0006, Version 1.2 /B06/;
- Applicable VCS tools, including VT0008: Additionality Assessment, Version 1.0 /B03/ and CDM TOOL30 v4.0 /B03/;
- Host country regulatory requirements, where applicable.

Scope of validation and verification includes:

- Validation of updated project documentation in alignment with VM0050 v1.0 requirements;
- Verification of proper application of revised baseline scenario, emission factors, and monitoring parameters;
- Assessment of the implementation and operational consistency with the PD /01/;
- Verification that monitored data are accurate, complete, traceable, and supported by objective evidence;
- Evaluation of the recalculated emission reductions for transparency, conservativeness, and eligibility for issuance.

Key phases of the engagement included:

1. Desk-based review of submitted documentation;
2. Interviews and information exchange with relevant stakeholders;
3. Resolution of clarifications and findings;

4. Issuance of the Final Validation and Verification Report (Requantification Verification report) and associated deliverables.

All activities were conducted in accordance with Carbon Check's internal quality assurance procedures, ensuring impartiality and independence throughout the engagement. No consulting or advisory services were provided to the Project Proponent in relation to this project. The verification process included appropriate consultation with the Project Proponent and relevant stakeholders, in line with the requirements of the VCS Standard; such consultation did not constitute the provision of consulting or advisory services.

1.3 Level of Assurance

The validation and requantification verification has been conducted with a reasonable level of assurance, in accordance with ISO 14064-3:2019 and the Verified Carbon Standard (VCS) Version 4.7 /B01/.

☒ Reasonable level of assurance

☐ Limited level of assurance

The materiality threshold applied for the identification of quantitative errors, omissions, or misstatements was established at one percent (1%) of the total reported GHG emission reductions, corresponding to 291.33 tCO₂. This threshold is determined in accordance with the requirements set out in Section 4.1.10 of the VCS Standard v4.7 /B01/.

To achieve this level of assurance, the Verification and Validation Body (VVB) applied a risk-based audit approach, including a targeted review of areas with potential material discrepancies associated with the updated methodology (VM0050 v1.0 /B02/) and the transition from the previously applied methodology (VMR0006 v1.2 /B06/).

The validation and verification activities involved:

- Review of the updated Project Description and requantification documentation;
- Cross-checking of calculation spreadsheets and monitored parameters;
- Recalculation and consistency checks of ex-post emission reductions;
- Interviews and clarifications with relevant project stakeholders.

Based on the verification activities undertaken and the evidence reviewed, Carbon Check concludes that the reported GHG emission reductions for the requantified monitoring period are free from material misstatements and in compliance with applicable VCS requirements and approved methodologies. The verification has been conducted independently and objectively, in accordance with Carbon Check's internal quality and impartiality procedures.

2 REQUANTIFICATION VERIFICATION PROCESS

2.1 Method and Criteria

Key Milestones	Date
Kick off Meeting	21-January-2025
Desk Review	25-June-2025 to 30-June-2025
On-site Inspection	01-July-2025 to 04-July -2025
Opening Meeting	01-July -2025
Closing Meeting	04-July -2025
DVR Issuance	15-July-2025
Technical Review	30-September-2025

The validation and requantification verification was conducted in three distinct phases to ensure methodological rigor, data integrity, and full alignment with the VCS Standard v4.7 /B01/ and ISO 14064-3:2019.

1. Desk Review and Completeness Check

A comprehensive desk-based review was carried out to assess the completeness, transparency, and consistency of all documentation submitted in support of the validation and requantification verification. This included:

- The revised VCS Project Description (PD) v1.8 /01/,
- The Requantification Report v1.2 for the first monitoring period (21-January-2023 to 23-August-2023),
- Revised GHG emission reduction calculation spreadsheets /02/;
- The previously applied methodology VMR0006 v1.2 /B06/,
- The updated methodology VM0050 v1.0 /B02/ and associated tools including CDM AMS-II.G v13.1 /B02/, CDM TOOL30 v4.0 /B03/, and VCS Tool VT0008 v1.0 /B03/.

Emphasis was placed on assessing the updated baseline scenario, demonstration of additionality, revised monitoring parameters, and recalculated emission reductions. The desk review also focused on the completeness and integrity of QA/QC procedures, especially in relation to parameters derived from analytical measurements and standardized testing. The outcome of this phase informed the subsequent risk assessment and sampling strategy.

2. Interviews and Evidence Verification

Following the desk review, the verification team designed and implemented a risk-based evidence-gathering plan. This phase involved:

- Verification of the alignment between on-ground implementation and the revised Project Description,
- Cross-verification of improved cookstove (ICS) distribution records /09/ and monitoring data,
- Review of updated emission reduction calculations /02/ and the application of new default values and emission factors,
- Evaluation of equipment calibration records /25/ and the QA/QC framework,
- Interviews with project stakeholders and monitoring staff to confirm procedural adherence and technical competence,
- Assessment of sampling methodologies applied in line with methodological requirements.

The purpose of this phase was to confirm that the monitoring system remains robust and that emission reductions have been calculated in a conservative and transparent manner under the new methodology.

3. Resolution and Final Reporting

Any gaps, inconsistencies, or discrepancies identified during the verification process were addressed through formal clarification requests and follow-up discussions with the Project Proponent. Upon resolution of all findings, the validation and verification team finalized its conclusions and issued this report in accordance with the VCS Standard, along with the signed VCS Verification Deed of Representation.

2.2 Document Review

The validation and requantification verification was conducted as a document-based audit to assess the accuracy, completeness, and consistency of the information provided by the Project Proponent, in accordance with the VCS Standard v4.7 /B01/ and the applied methodology VM0050 v1.0 /B02/. Carbon Check carried out a comprehensive review of the revised VCS PD version 1.8, dated 02-January-2026 /01/, revised GHG emission reduction calculation spreadsheets /02/; the requantification Report version 1.2, dated 02-January-2026 /03/, and all relevant supporting documentation submitted by the Project Proponent.

The review included a cross-check of reported monitoring data against primary sources such as the improved cookstove (ICS) distribution database /09/, stove tracking records, instance 01 KPT result database /22/, instance 01 f_{NRB} calculation database /20/ and user monitoring survey results /08/. Specific attention was paid to the implementation of the updated monitoring plan outlined in the revised PD, particularly regarding the frequency and method of data collection, calibration and maintenance of monitoring equipment, and the application of QA/QC procedures as per the revised methodology.

Emission reduction calculations, default values, and input assumptions used in the recalculated ex-post spreadsheets were evaluated for methodological consistency, transparency, and conservativeness. The verification team ensured that both analytically derived and standardized test parameters adhered to the quality requirements set forth under VM0050 v1.0 /B02/ and its associated tools.

Where discrepancies, data gaps, or methodological deviations were observed, Corrective Action Requests (CARs) and Clarification Requests (CLs) and a Draft Verification Report (DVR) was also issued. These were addressed by the Project Proponent, and revised documents were submitted for reassessment. Final conclusions were drawn after a detailed review of the updated documents and supportive evidence. A full list of reviewed documents and references is included in Appendix 2.1.

2.3 Interviews

As part of the validation and requantification verification process, an on-site assessment of the project activity was conducted from 01-July -2025 to 04-July -2025. The primary objective of the visit was to verify the continued implementation and operation of the project in accordance with the revised PD, the applied methodology VM0050 v1.0 /B02/, and the requirements outlined in the VCS Standard v4.7 /B01/.

Structured interviews were conducted with key project personnel involved in project implementation, monitoring, data management, and quality assurance. These interviews served to:

- Confirm the alignment of field operations with the updated monitoring plan and methodology requirements,
- Validate data collection processes, QA/QC procedures, and the accuracy of emission reduction calculations,
- Clarify elements of the revised PD and address questions or discrepancies raised during the desk review, and
- Assess updates to operational practices and monitoring systems following the transition to VM0050 v1.0.

The personnel interviewed and the main topics discussed are summarized below:

Sl. No	Date	Name	Organisation	Topic	Interviewed by
1	01-July - 2025 to 04-July - 2025	Dr. Ramanan Loganathan	Tramontana	Interviews and field discussions were conducted to evaluate the project's performance against the approved PD, Monitoring Report, and applied methodology for both instances. Key topics included: - Project implementation and alignment with the PD in Instance 2. - ICS distribution status, operational performance, and usage patterns. - Emission reduction monitoring, including data collection, accuracy, and consistency with methodology. - Baseline scenario validation for Instance 2, supported by field data and KPTs. - Monitoring framework, including data flow, documentation, and household-level verification. - Verification of monitored parameters, including sampling methods, KPT data, and ER calculations as per VM0050. - Verification of SDG contributions, including reduced fuelwood use, health improvements, and time savings. - Assessment of "No Net Harm" compliance, with	Manas Hadler and Nisha Jha
2	01-July - 2025 to 04-July - 2025	Jessica Dring-Morris	Tramontana		
3	01-July - 2025 to 04-July - 2025	Saibaba Abburi	Tramontana		
4	01-July - 2025 to 04-July - 2025	Narasimha Barpata	Tramontana		

				documented mitigation for identified risks. • Legal ownership and entitlement to carbon credits, verified through declarations and legal documentation. • Environmental and social co-benefits, confirmed through stakeholder interviews. • Stakeholder engagement and community participation, including meeting records and community feedback. • Grievance redress mechanism, reviewed for accessibility, transparency, and any recorded grievances. • Governance and safeguards, ensuring compliance with national regulations and project-specific commitments. • Verification of control households, confirming stability of baseline conditions and absence of leakage or project interference.	
5	02-July - 2025	Vanthal Bangaru Naidu	Tramontana (Mandal Coordinator)	• Standard operating process • Baseline survey • Distribution • Maintenance and grievance redressal • Monitoring survey • KPT • Data recording and QA/QC procedure • Training	
6	03-July - 2025	Thodomu Mohan Rao	Tramontana (Mandal Coordinator)	• Standard operating process • Data recording and QA/QC procedure • Training	
7	03-July - 2025	Jabbadu Dhanalakshmi	Tramontana (Computer Operator)	• Standard operating process • Data recording and QA/QC procedure • Training	

8	03-July - 2025	L. Dhanurajo	Tramontana (Maintenance Head)	- Standard operating process - Maintenance and grievance redressal - Data recording and QA/QC procedure - Training	
9	09-July - 2025 (Online)			- Product design - R&D - Technical parameters including efficiency - Supply agreement - Customer support - Training	
10	01-July - 2025	Majji Suramma	End user (Instance 1)	- Baseline scenario - Baseline survey and KPT	
11	01-July - 2025	Gudiya Nageswara Rao	End user (Instance 1)		
12	01-July - 2025	Gollori Prahladh	End user (Instance 1)		
13	02-July - 2025	Vanthala Aso	End user (Instance 1)		
14	02-July - 2025	Masadi Kundiramma	End user (Instance 1)		
15	02-July - 2025	Masadi Nagaraju	End user (Instance 1)		
16	02-July - 2025	Seedari Padma	End user (Instance 1)		
17	02-July - 2025	Boyini Eswaramma	End user (Instance 1)		
18	03-July - 2025	Korra Appanna	End user (Instance 1)		
19	03-July - 2025	Majji Balaraju	End user (Instance 1)		
20	03-July - 2025	Pangi Pentamma	End user (Instance 1)		
21	01-July - 2025	Killo Chinni	Control Household (Instance 1)		
22	02-July - 2025	Vanapala Appalakondamma	Control Household (Instance 1)		

23	02-July - 2025	Korabu Savitramma	Control Household (Instance 1)		
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To supplement the validation and verification of monitored data and emission reductions, To supplement the validation and verification of monitored data and emission reductions, the VVB applied the CDM Sampling and Surveys for CDM Project Activities and Programmes of Activities, Version 09 /B04/ with an acceptance sampling plan using an Acceptance Quality Limit (AQL) of 0.5%, Upper Quality Limit (UQL) of 20%, producer/consumer risks of 10% each, and an acceptance number (c) of 0. A total of 11 samples were initially selected, and oversampling was conducted by adding 3 additional households, reaching 14 in total.

These households were chosen to ensure representative coverage of stove usage patterns and variations in user behavior. On-site inspections were carried out at each of the selected households in accordance with the requirements of VM0050 v1.0 /B02/.

The inspections and interviews were conducted to assess:

- The correct installation and operational condition of improved cookstoves (ICS),
- User awareness and training related to stove usage and maintenance,
- Consistency with monitoring records and reported usage behavior, and
- Applicability of methodology conditions, including eligibility criteria and performance benchmarks.

These interviews formed a key component of verifying that the reported GHG emission reductions are based on credible, traceable, and conservatively monitored data. The verification team ensured that the project continues to conform to the requirements set forth in the revised PD and applicable methodology.

2.4 Site Visits

A site visit was conducted by the verification team from 01-July-2025 to 04-July-2025 in accordance with the VCS Standard v4.7 /B01/ and the applied methodology VM0050 v1.0 /B02/. The purpose was to verify the implementation of the project activity, assess monitoring practices, and confirm consistency with the revised Project Description (PD) /01/.

Field inspections were carried out in existing specifically the Paderu division of Alluri Sitarama Raju district, Andhra Pradesh. Fourteen households were selected using a risk-based sampling approach to ensure representative coverage.

On-site activities included:

- Verification of stove distribution, installation, and operational use;
- Validation of PD and QA/QC procedures;
- Review of training records, distribution logs, and field documentation;
- Interviews with end-users and project personnel.

Sampling approach

The total population of the Improved Cookstoves (ICS) distributed under the project activity until the end of this monitoring period dated 23-August-2021 is 46,160 (Instance 1).

The sampling plan implemented by the Project Proponent (PP) is in accordance with the applied approved monitoring methodology VCS Methodology VM0050, Version 1.0 /B02/ and the revised Project Description (PD) /01/, and the CDM Standard: Sampling and Surveys, Version 09.0 /B04/.

The PP employed Simple Random Sampling in line with VM0050 v1.0 /B02/ and the registered monitoring plan. Sampling activities—including household surveys, Kitchen Performance Tests (KPTs), were carried out by trained field teams using the Tramontana ICS software. Each stove carried a unique serial number, ensuring traceability.

The following parameters are not calculated on a sample basis, but instead on the total project population and do not form part of sample-based monitoring:

- $N_{j,k,y}$: Number of commissioned devices of type j from batch k in year y
- n_{HH} : Number of ICS distributed to each household

For the above parameters VVB randomly selected 96 samples from the database and performed full data check at higher data aggregation levels and sampling at original data levels (based on 95/10 confidence / precision and 50 % response distribution¹)

The following monitoring parameters were assessed through surveys and testing:

- $\eta_{j,k,y}$: Proportion of ICS still regularly used
- $BC_{b,i,y}$: Average baseline fuel consumption
- $BC_{p,j,k,y}$: Average project fuel consumption
- $Hh_i / Hh_{j,k}$: Household size under baseline/project conditions
- $\eta_{new,avg,y}$: Weighted average efficiency of project stoves (via WBT)

¹ <http://www.raosoft.com/samplesize.html>

For parameters $n_{j,k,y}$ and $n_{new,avg,y}$, the values were already determined via sampling survey and measurement on project devices during monitoring activity performed by the PP before joint validation and verification for MP1 as $n_{y,i,j}$ and $n_{new,s}$ respectively. Since these values are still hold good as per the revised PD, the VVB does not require to re-verify these via applying any sampling methods.

For the other parameters namely, $BC_{b,i,y}$, $BC_{p,j,k,y}$, and $Hh_i / Hh_{j,k}$, the PP conducted 151 baseline survey (in additional to the 1793 baseline surveys already performed before registration) and KPTs on 151 baseline devices on the same households (including 50 control households) and KPTs on 100 project devices on the same non-control households. VVB applied the CDM Sampling and Surveys for CDM Project Activities and Programmes of Activities, Version 09 /B04/ with an acceptance sampling plan using an Acceptance Quality Limit (AQL) of 0.5%, Upper Quality Limit (UQL) of 20%, producer/consumer risks of 10% each, and an acceptance number (c) of 0. A total of 11 samples were initially selected, and oversampling was conducted by adding 3 additional samples, reaching 14 in total.

The VVB verified ICS usage and distribution records—no discrepancies were observed (i.e., $c = 0$), and the results were accepted. From a population of 151 households, consisting of 100 non-control households common for both baseline survey and baseline and project KPT plus 50 control households exclusively for baseline survey and baseline KPT, the VVB reviewed household-level baseline scenario and cooking pattern and baseline and project KPT data, including fuel quantities, usage days, refuelling frequency, and household size. All values aligned with records captured by the PP.

The population sizes and applied VVB sample sizes are detailed below:

Parameter	Verification Approach	Population ²	VVB's Sample Size
$N_{j,k,y}$ Number of commissioned devices of type j from batch k in year y	COM	62,121	96
n_{HH} Number of ICS distributed to each household			
$n_{j,k,y}$ Proportion of commissioned devices of type j from batch k still regularly used in year y	-	-	-

² Only instance 1 population is applicable, since instance 2 was not started during MP1

$BC_{b,i,y}$ Average fuel quantity per baseline device type i in year y	ASP	151	14 ³
$BC_{p,j,k,y}$ Average quantity of fuel used by project device type j from batch k during year y	ASP	100	11
$Hh_i / Hh_{j,k}$ Average household size of the target population using baseline device type i Average household size of the target population using project device type j from batch k	ASP	151 / 100	14 ⁴ / 11
$\eta_{new,avg,y}$ Weighted average efficiency of project devices in year y	-	-	-

Compliance with Sampling Standard /B04/

The VVB confirms that all monitored values were supported by verifiable documentation and field evidence. The PP's sampling design was reviewed and found to be in compliance with **Paragraph 33 of the CDM Standard: Sampling and Surveys, Version 09.0/B04/**, which states:

Paragraph 33:

"The DOE shall assess the sampling plan applied by the CME or the PP against the sampling requirements of this standard and, where applicable, against the registered PD or PoA-DD, and shall apply an independent sampling effort to verify the reliability of data and information collected."

Additionally, the verification adhered to **Paragraphs 25 to 28** of the same standard:

Paragraph 25:

"The DOE shall verify that the CME or the PP has implemented the sampling and survey process in accordance with the applied methodology, methodological tool, and registered PD or PoA-DD."

³ Required sample size was 11, with oversampling conducted by adding 3 additional samples, total samples reached 14

⁴ Required sample size was 11, with oversampling conducted by adding 3 additional samples, total samples reached 14

Paragraph 26:

“If the CME or the PP has adopted a sampling-based approach, the DOE shall also apply a sampling approach for the purpose of verification. The DOE shall include in the verification report the size of the population subject to verification, sample size, sampling method used, confidence/precision achieved, and details of the sample.”

Paragraph 27:

“The DOE shall check that the size of the sample used by the CME or the PP is sufficiently large to achieve the desired precision level for each parameter based on the applied sampling method.”

Paragraph 28:

“Where the CME or the PP uses sampling, the DOE shall apply acceptance sampling for the purpose of verification.”

The VVB reviewed the PP’s sampling methodology, sample size calculations, and survey implementation, and confirms that the required **90/10 confidence/precision level** was met for all parameters. No deviations were identified.

The sampling and verification were found to be compliant with the following documents:

/B01/ VM0050 Methodology for Improved Cookstoves v1.0

/B02/ VCS Standard v4.4

/B03/ VCS Methodology Change and Requantification Procedure v4.0

/B04/ CDM Standard: Sampling and Surveys, Version 09.0

/B06/ Clean Cooking Alliance’s WBT Protocol v4.2.3 (2014)

The VVB concludes that the sampling approach adopted by the Project Proponent is statistically valid, transparent, and consistent with both the applied methodology and the relevant standards.

The site visit confirmed the accuracy of field implementation and monitoring arrangements, supporting the credibility of the reported emission reductions for the monitoring period from 21-January-2023 to 23-August-2023. Verification was concluded with a reasonable level of assurance.

Activity	Place	Topics
Opening meeting	Project Site	Brief description of project activity and scope of audit.

Interviews with representatives of PP including distribution/installation, implementation and monitoring/testing teams	Project Site / PP's Local Office in Andhra Pradesh	• Implementation status • ICS distribution • QA/QC process • Database management • Trainings conducted • On-going stakeholder consultation • Grievance redressal • Carbon credits ownership rights • Project documentation • Emission reduction calculations • Supporting evidence
Interview with ICS manufacturing partner	Online	• Product design • R&D • Technical parameters including efficiency • Supply agreement • Customer support • Training
Interviews with baseline stove users / end-users	Project Site	• Cross-check PP's <i>baseline</i> survey/KPT records
Closing meeting	Project Site	• Discussion on findings

2.5 Resolution of Findings

This section summarizes the outcomes of the validation and requantification verification process, including the document review, on-site assessments, and interviews with project personnel and beneficiaries.

During the course of the verification, the verification team identified a total of **08 Corrective Action Request (CARs)** and **06 Clarification Requests (CLs)**, and **02 Forward Action Requests (FARs)** were raised in this verification, and there were **02** outstanding **FARs** from previous assessments.

The findings were issued to address discrepancies, information gaps, or potential non-conformances in relation to the applied methodology VM0050 v1.0 /B02/, the revised Project Description (PD), and the requirements of the VCS Standard v4.7 /B01/.

Validation and Verification Finding Categories

- **Corrective Action Request (CAR):** Issued when a material issue was found that could affect the accuracy of reported emission reductions or indicated non-compliance with methodological or VCS requirements.
- **Clarification Request (CL):** Raised where further information or clarification was needed to confirm conformity or enhance transparency.

Each CAR and CL was formally communicated to the Project Proponent. The proponent responded with the necessary documentation, clarifications, and updates to the PD and requantification report and supporting spreadsheets where applicable. The verification team reviewed these responses and determined that the actions taken were sufficient to close the findings.

The resolution of the CAR led to minor revisions in the PD and Requantification documentation, specifically ensuring the consistency and completeness of parameter reporting. CLs mainly addressed the transparency of monitoring procedures, stove usage documentation, and data traceability. No changes to the estimated emission reductions were required as a result of the findings.

A detailed summary of each CAR and CL, along with the issues raised, responses provided by the Project Proponent, and the conclusions drawn by the verification team, is presented in **Appendix 5**.

2.5.1 Forward Action Requests

The two Forward Action Requests (FARs) identified during the previous joint validation and verification have been satisfactorily resolved in the course of this verification. In addition, two new FARs have been raised as a result of the current verification, which are to be addressed in subsequent audits. A detailed summary of the FARs raised during this verification is provided in Appendix 5.

3 VALIDATION FINDINGS

3.1 Project Details

The verification team has assessed the PD /01/ and confirms that it is accurate, complete, and provides a clear understanding of the project's nature, including its objectives, scale, location, technology type, implementation framework, and expected environmental and social impacts.

The PD is consistent with the applied methodology (VM0050 v1.0) and the requirements of the VCS Standard v4.7 /B01/. The description outlines the baseline conditions, project scenario, and monitoring plan in a manner that is transparent, technically sound, and verifiable. The project boundaries, duration, and crediting period have also been correctly identified. No inconsistencies or omissions were observed that could materially affect the emission reduction calculations or the eligibility of the project under the VCS Program.

Item	Evidence gathering activities, evidence checked, and assessment conclusion
General eligibility of the project to participate in the VCS Program	- The project involves the distribution of energy-efficient cookstoves under Sectoral Scope 03 – Energy Demand. The applied methodology, VM0050 v1.0 (<i>Methodology for Improved Cookstoves and Thermal Energy Devices</i>) /B02/, is approved under the VCS Program and applicable to the project type and scale. - As confirmed in Section 1.4.1 of the revised Project Description (PD) /01/ and in line with Table 2.1 of the VCS Standard v4.7 /B01/, the project does not fall under any exclusion criteria. It remains eligible under the scope of the VCS Program. - The project was listed on the Verra Registry on 16-June-2023 and is being transitioned from the previous methodology VMR0006 v1.2 to the updated VM0050 v1.0 (published October-2024), in accordance with current program requirements. - As per Section 3.6.8 of the VCS Standard v4.7, all project instances within ten kilometres under the same proponent are included as part of this single project. No evidence of disaggregation or artificial fragmentation was identified. - Based on the review, the verification team confirms the project's continued eligibility under the VCS Program and

	compliance with all relevant requirements of the VCS Standard and applied methodology.
Not applicable as the proposed project activity is a non-AFOLU project.	Not applicable as the proposed project activity is a non-AFOLU project.
Requantification eligibility	The project proponent has applied the Methodology Change and Requantification Procedure v4.0 /B01-f/ in accordance with the requirements of the VCS Standard v4.7/B01/. Based on the verification team's assessment, the project meets all eligibility conditions to apply this procedure, as outlined below: - The project meets all applicability conditions of the new methodology, VM0050 v1.0 (Methodology for Improved Cookstoves and Thermal Energy Devices), which supersedes the previously applied methodology (VMR0006 v1.2) /B06/. The methodology switch has been fully applied to the project in accordance with Section 1.1.2 of the Procedure /B01-f/. - The project proponent has maintained complete and reliable datasets aligned with the parameters, data requirements, and calculation procedures specified in VM0050 v1.0. No methodology deviations were applied to compensate for missing data. - The monitoring period being requantified (21-January-2023 to 23-August-2023) is the first monitoring period of the project and is the latest Verra-approved monitoring period at the time of this verification. No gaps exist between the reported verification periods. - Verified Carbon Units (VCUs) from the previous monitoring period, subject to the joint validation and verification, have been issued and are currently active on the Verra Registry. In accordance with Section 1.1.2 of the Requantification Procedure v4.0 /B01/, the project proponent has implemented the following permitted changes: - Full application of the new methodology (VM0050 v1.0); - Reassessment of the baseline scenario where applicable;

	• Update of the monitoring approach, including data parameters and QA/QC procedures; • Requantification of GHG emission reductions for the affected monitoring period. The methodology switch has been formally notified to Verra, and the revised PD reflects the updated methodological and monitoring framework. Based on the review of all submitted documents and evidence, the verification team concludes that the project is eligible to apply the Methodology Change and Requantification Procedure v4.0 /B01/.
Project scale and estimated ERRs	The project remains within the originally registered scale and continues to qualify as a large-scale activity, in accordance with Section 1.11 of the revised Project Description (PD) /01/ and § 3.10.1 of the VCS Standard v4.7. A total of 46,160 improved cookstoves (ICS) were distributed and monitored during the requantified monitoring period (21-January-2023 to 23-August-2023), consistent with the registered capacity and implementation boundaries. The estimated emission reductions (ERRs) for the requantified period have been revised to 29,133 tCO₂e. This represents a reduction from the previously verified figure of 62,743 tCO₂e, resulting from the updated calculation procedures and default parameters introduced under the applied methodology VM0050 v1.0 /B02/, including changes to baseline fuel consumption and the fNRB. The updated annual average emission reductions over the first crediting period are estimated at 768,748 tCO₂e. These revisions have been applied in accordance with the VCS Requantification Procedure v4.0 /B01/ and reflect full implementation of the new methodology.
Likelihood of achieving estimated ERRs	The VVB has reviewed the revised emission reduction estimates presented under VM0050 v1.0 /B02/ and assessed their consistency with field monitoring data, stove usage surveys, and QA/QC documentation. The emission reduction calculations were independently reproduced and cross-verified against the monitoring database and applied assumptions.

	The assumptions related to stove usage rates, fuel savings, and displacement of baseline technologies were found to be aligned with field observations and comply with the methodological requirements. Based on this review, the VVB confirms that the project activity is likely to achieve the estimated GHG emission reductions for the requantified monitoring period.
Conditions prior to project initiation	There have been no modifications to the baseline scenario or associated parameters during the current verification. The baseline remains valid and appropriately reflects pre-project cooking practices. Baseline data remains valid and appropriately reflects pre-project cooking practices. Households were previously reliant on traditional biomass stoves using firewood as the dominant fuel type, with inefficient thermal performance. There have been no updates to the baseline scenario during this verification.
Additional information relevant to the project	No commercially sensitive or proprietary information has been excluded from public versions of project documentation. The verification team confirms that the project is in compliance with the VCS Standard Section 3.5.2 /B01/ regarding public disclosure. There have been no material changes in project stakeholders or implementing entities since project registration or during the current monitoring period. This is in compliance with Section 4.1.17 of the VCS Standard v4.7 /B01/. QA/QC procedures have been reviewed and found to be sufficient and compliant with the requirements under VM0050 v1.0. Relevant adjustments have been made to ensure that data quality and traceability are maintained in the context of the methodology transition.

3.2 Application of the New Methodology for the Requantification

3.2.1 Title and Reference

Type	Reference ID	Title	Version
Methodology	VM0050	Energy Efficiency and Fuel-Switch Measures in Cookstoves ⁵	1.0
Methodology	CDM AMS-II G	Energy efficiency measures in thermal applications of non-renewable biomass ⁶	13.1
VCS Tool	VT0008	Additionality Assessment ⁷	1.0
CDM Tool	CDM TOOL30	Calculation of the fraction of non-renewable biomass	4.0

3.2.2 Applicability

Methodology ID	Applicability condition	Assessment and conclusion
VM0050	The project activity corresponds to: a) Replacement of non-renewable biomass (e.g. firewood, charcoal)-fired cookstoves with any of the following:	Verified through review of the Project Description /01/ and supporting documentation /02/-/33/. The project replaces traditional wood-fired cookstoves with improved cookstoves using the same fuel type. Conforms with condition a)(i).

⁵ <https://verra.org/wp-content/uploads/2024/10/VM0050-EE-and-Fuel-Switch-Measures-in-Cookstoves-v1.0.pdf>

⁶ <https://cdm.unfccc.int/UserManagement/FileStorage/6TOUCX21D0BHNVRZFWMEKALY94GS7>

⁷ <https://verra.org/wp-content/uploads/2024/10/VT0008-Additionality-Assessment-v1.0.pdf>

	(i) More efficient project devices that use the same fuel as the baseline; (ii) Efficient project devices fired by renewable biomass or bioethanol; (iii) Efficient project devices fired by liquefied petroleum gas (LPG); or (iv) Electric-powered project devices b) Replacement of solid or liquid fossil fuel (e.g. coal, kerosene)-fired cookstoves with any of the following: (i) Efficient project devices fired by renewable biomass or bioethanol; (ii) Efficient project devices fired by LPG; or (iii) Electric-powered project devices	
VM0050	Project devices are used in households, communities, institutions or small or medium enterprises (SMEs), collectively referred to in this methodology as the “target population”	The project activity is implemented in households located in rural areas of the ASR district of Andhra Pradesh in the first instance. The assessment team verified the implementation of the activity through direct observation during the onsite visit /19/, where improved cookstoves were found installed and operational in beneficiary households. The distribution records /09/ reviewed and the data captured through the Tramontana ICS software further

		confirmed household-level deployment. Therefore, this applicability condition is met.
VM0050	Project devices using renewable biomass (fuel-switch) or non-renewable biomass (improved efficiency) are single-pot, multi-pot portable, or in-situ cookstoves with an initial thermal efficiency of at least 25%	The Tramontana ICS units installed under the project activity are high-quality single-pot portable cookstoves designed for use with woody biomass. Independent thermal efficiency testing confirms that the CRS-23 and STAR-25 models achieved efficiencies of 36.01% and 52.50%, respectively, as verified from the thermal efficiency test report /21/. Both values exceed the 25% threshold required by the methodology. Therefore, this applicability condition is fully met.
VM0050	The project proponent designs incentive mechanisms to reduce the use of inefficient baseline devices and practices that can be replaced by the project devices and describes these mechanisms in the project description	End-user license agreements /26/ include provisions for removal of baseline stoves. Field staff conduct follow-up visits and a grievance mechanism is in place. Condition is met.
VM0050	Where a project device has ended its technical life, the project proponent either replaces it with a comparable or better project device or retrofits its essential components to continue meeting the minimum service level requirements (i.e., thermal energy generation), otherwise no further emission reductions may be claimed for the project device	Project documentation confirms commitment to repair, retrofit, or replace ICS after end of life. A tracking system for replacements is in place and verifiable. Applicability condition is met.
VM0050	Project proponents implement a method for the distribution and identification of project devices that avoids double counting of emissions reductions by other mitigation	The VVB assessed the project's stove distribution and identification system to determine whether it effectively prevents double counting of emission reductions. Each ICS distributed under

	actions and includes unique product identification on the stove itself at the time of distribution/sale (e.g., program logo, alpha/numeric ID, and end-user location, such as geographical coordinates, complete address information)	the project is uniquely identifiable through a serial number and QR code, together with the Project Proponent's logo, permanently affixed to the stove. These identifiers are digitally linked to individual households via the Tramontana ICS software, which records beneficiary information, installation date, and geolocation data. The VVB verified the correct implementation and functionality of this system through a review of the distribution database /09/ and onsite observations /19/. In addition, the VVB reviewed the Project Description and conducted checks of Verra and other relevant carbon registries to confirm that no other carbon cookstove projects or schemes are operating within the project boundary. Household survey records and onsite interviews further confirmed that beneficiary households did not possess efficient cookstoves associated with other carbon projects or schemes. Based on this assessment, the VVB concludes that the project implements an adequate and reliable identification and tracking system, and that the risk of overlaps or double claiming of emission reductions due to stoves from different carbon projects or schemes is effectively mitigated. Therefore, this applicability condition is fully met.
VM0050	The project complies with any national, sub-national, or local regulations or guidance for the installation, commercialisation, distribution and use of improved cookstoves and/or fuel supply and	Based on the review of applicable national, sub-national, and local regulatory frameworks, as well as confirmation from project stakeholders, there are no mandatory laws or regulations in the host country

	use for the target population. National, regional, and local regulatory frameworks for the provision of the type of thermal energy services provided by the project activity must be documented	that require or restrict the distribution, installation, or use of improved cookstoves. The project is implemented on a voluntary basis and does not contravene any legal requirements. Therefore, this applicability condition is fully met.
VM0050	Where project activities reduce emissions from non-renewable biomass, including firewood and charcoal, the risk of double counting is assessed on a national basis by evaluating at validation and crediting period renewal whether there are REDD+ projects or jurisdictional REDD+ programs whose boundary overlaps with the expected fuel source area of the project. The project proponent must report on the findings of this assessment for informational purposes in the project description	The project proponent provided a documented assessment demonstrating that no REDD+ or jurisdictional REDD+ projects overlap with the expected biomass sourcing area. This was substantiated by the verification team's independent review of relevant carbon credit registries. As this requirement is for informational purposes only under the methodology, the condition is considered met.
CDM AMS-II G	The aggregate energy savings of a single project activity shall not exceed the equivalent of 60 GWh per year or 180 GWh thermal per year in fuel input	The emission reductions spreadsheet /02/ submitted by the project proponent confirms that the energy savings from the project activity remain below the 60 GWh per year threshold, as required under AMS-II.G. The verification team has reviewed and validated the underlying calculation logic and assumptions. This applicability condition is therefore satisfied.
CDM AMS-II G	Non-renewable biomass has been used in the project region since 31 st December 1989, using survey methods or referring to published literature, official reports or statistics	The project proponent has submitted supporting evidence from recognized national and international sources confirming long-term use of non-renewable biomass in the project region. The VVB has reviewed Table 8 of the UN FAO report, which shows

		significant firewood demand in India between 1970 and 1995. Additionally, State of Forestry reports from the Forest Survey of India indicate substantial forest degradation between 1993 and 1997, with 671,000 hectares of forest lost and 1,831,000 hectares degraded. Further, Global Forest Watch data confirms a 6% tree cover loss in India from 2001 to 2023. Based on this documentation and analysis, the VVB confirms that non-renewable biomass has been in use since at least 31 December 1989. This applicability condition is therefore met.
CDM AMS-II G	For cases where the biomass is sourced from renewable sources, the project participants should use a corresponding Type I methodology	Not applicable to this project. Biomass used is non-renewable. No compliance action required.
CDM AMS-II G	The VCS project document shall also explain how the proposed procedures prevent double counting of emission reductions, for example, to avoid that project stove manufacturers, wholesale providers or others claim credit for emission reductions from the project devices	The VVB reviewed the signed license agreements /26/ provided by the Project Proponent and confirms that the rights to all emission reductions generated under the project are explicitly and solely assigned to the Project Proponent. No evidence of any third-party claims or conflicting ownership was identified. This satisfies the applicability condition related to the exclusive ownership of emission reductions. Furthermore, each improved cookstove (ICS) distributed under the project is clearly marked with a unique serial number and QR code, along with the Project Proponent's logo. This identification system is integrated with the Tramontana ICS

		software application, which captures and links detailed beneficiary data, including personal information, address, geolocation, and installation date. The VVB verified the accuracy and presence of these unique identifiers through review of the distribution database /09/ and during the onsite inspection /19/. Therefore, the applicability condition related to the prevention of double counting is fully met.
VT0008	This tool is applicable under any of the following conditions: 1. The applicable methodology requires or permits the use of this tool; or The VCS Program rules and requirements require or permit the use of this tool	The project's use of VT0008 for baseline scenario assessment is consistent with the requirements of methodology VM0050. The VVB reviewed the project documentation and confirms that VT0008 /B03/has been correctly applied in accordance with the methodology guidance. The use of this tool is appropriate and justified for the project context. Therefore, this applicability condition is fully satisfied.
CDM TOOL30	This tool may be used by: a) DNAs to submit region- or country-specific default fNRB values, following the procedures for development, revision, clarification and update of standardized baselines (SB procedures); or project participants to calculate project- or PoA-specific fNRB values.	The project applies TOOL30/B03/ for the calculation of project-specific fNRB /20/ values in line with methodological requirements. The VVB has reviewed the calculations and supporting documentation and confirms that TOOL30 has been correctly and consistently applied. Consequently, this applicability condition is fully met.

3.2.3 Project Boundary

The VVB confirms that the project boundary for the grouped activity is defined as the geographical boundary of India, within which all project activities are implemented. Each project instance maintains a clearly defined and distinct boundary to enable localized monitoring, verification, and emission accounting. For instance 1, the project boundary corresponds to the Paderu division of the Alluri Sitarama Raju district, Andhra Pradesh, with approximate coordinates of latitude 18.011959 and longitude 82.555030.

Validation of the project boundary was undertaken through a review of project documentation, including the Project Description (PD), commissioning reports, and spatial data. A site inspection was conducted within the defined boundary to confirm the operational presence of project activities, verify beneficiary locations, and ensure alignment with the boundaries described in the PD.

In accordance with the VM0050 v1.0 /B02/ methodology requirement for projects involving non-renewable biomass fuels, the Project Proponent has provided a KML file /27/ delineating the geographic area(s) of origin of the biomass fuel used under the project along with a description of how these areas are defined. The VVB reviewed the KML file and confirmed that it is consistent with the boundary description provided in the PD and with the fuel sourcing practices verified during the site visit.

The VVB concludes that the project boundary has been appropriately defined and validated.

Identification and Selection of GHG Sources, Sinks, and Reservoirs

The VVB verified that all relevant greenhouse gas (GHG) sources and gases are identified and appropriately included or excluded from the project boundary in line with Section 5 and Table 1 of VM0050 v1.0. The primary source of emissions under both the baseline and project scenarios is thermal energy generation from biomass combustion in cookstoves.

The table below outlines the inclusion of relevant GHG sources and gases within the project boundary:

Source		Gas	Included?	Assessment and conclusion
Baseline	Thermal energy generation	CO ₂	Yes	Emissions from combustion of non-renewable biomass in traditional cookstoves represent the primary source of baseline emissions. The VVB verified inclusion based on baseline household survey results /23/ and Kitchen Performance Test (KPT) data /22/ demonstrating exclusive use of

Source		Gas	Included?	Assessment and conclusion
Project	Production of charcoal fuel			firewood for cooking within the project area.
		CH4	Yes	Methane emissions from incomplete combustion of firewood in traditional cookstoves are material and correctly included. The VVB confirmed relevance through baseline survey findings and methodological consistency with VM0050 Section 5 and Table 1 /B02/.
		N2O	Yes	Nitrous oxide emissions from biomass combustion are conservatively included in the baseline boundary. The VVB verified consistency with VM0050 requirements and baseline combustion practices documented in the survey database /23/.
		CO2	No	Charcoal production does not occur in the project area. The VVB confirmed that baseline households rely exclusively on raw firewood and do not engage in charcoal production, based on baseline survey responses /23/ and onsite observations /19/.
		CH4	No	Not applicable, as charcoal production activities are absent within the defined baseline boundary, as verified through surveys /23/ and field interviews /19/.
		N2O	No	Not applicable due to the absence of charcoal production, as confirmed during baseline assessment /23/.
Project	Thermal energy generation	CO2	Yes	Emissions from biomass combustion in improved cookstoves remain the primary project-scenario emission source. The VVB verified inclusion based on monitoring records /08/ and confirmation that the project introduces biomass-fired improved cookstoves only /01/.
		CH4	Yes	Methane emissions from improved cookstove use are relevant and

Source	Gas	Included?	Assessment and conclusion
			conservatively included. The VVB confirmed consistency with VM0050 Section 5 and monitoring data /08/.
	N ₂ O	Yes	Nitrous oxide emissions from biomass combustion under project conditions are appropriately included. The VVB verified alignment with VM0050 requirements and project monitoring data /08/.
	CO ₂	No	Fuel transportation emissions are excluded as firewood is sourced locally from nearby forests without mechanized transport. The VVB confirmed this through onsite verification /19/ and beneficiary interviews.
	CH ₄	No	Not applicable due to absence of fuel transport activities, as confirmed during onsite assessment /19/.
	N ₂ O	No	Not applicable.
	CO ₂	No	No fuel processing or production activities occur within the project boundary. The VVB verified this through PD review /01/ and onsite inspection /19/.
	CH ₄	No	Not applicable.
	N ₂ O	No	Not applicable.
	CO ₂	No	The project does not involve any self-generated or auxiliary electricity systems. The VVB confirmed this through PD review /01/ and field verification /19/.
	CH ₄	No	Not applicable
	N ₂ O	No	Not applicable

Source	Gas	Included?	Assessment and conclusion
Grid electricity generation and distribution	CO ₂	No	Grid electricity is not used or generated under the project activity. The VVB verified exclusion based on PD Section 3.3 /01/ and onsite assessment /19/.
	CH ₄	No	Not applicable
	N ₂ O	No	Not applicable

3.2.4 Additionality

The project applies the Activity Method as outlined in the applied methodology VM0050 to demonstrate additionality. This approach involves two key steps: (i) Regulatory Surplus, and (ii) Positive List eligibility. As the project satisfies the requirements of both steps, it is deemed additional under the methodology. Consequently, the Project Method step—comprising barrier, investment, or common practice analysis—is not applicable.

To validate regulatory surplus, the VVB confirmed that the project is implemented in a UNFCCC Non Annex 1 country (India), where no laws, statutes, or regulatory frameworks exist that mandate the use or distribution of improved cookstoves. Furthermore, no systematically enforced regulations at the national or sub-national level require the adoption of efficient cookstove technologies. This was established through a review of relevant policy and legal documents, evaluation of stakeholder declarations, and interviews conducted during the site visit. The VVB also verified that the project activity is implemented on a voluntary basis, further supporting the regulatory surplus claim.

For positive list eligibility, the VVB reviewed the applicability conditions specified in VM0050 v1.0 /B02/ and confirmed that the project fulfills both criteria. First, the project introduces efficient biomass-fired cookstoves that replace traditional, inefficient biomass-fired cookstoves. This was substantiated through technical documentation of the cookstove models /16/ and distribution records /09/. Second, the project distributes the improved cookstoves at no cost to beneficiaries, and the only source of revenue is the sale of verified carbon units (VCUs). The VVB validated this through a review of end user license agreement section 1a /26/, beneficiary declarations, and the project's design document /01/. As both conditions are met, the project qualifies under the positive list.

Cross-checking of data was performed by the VVB through a combination of document reviews, field observations, stakeholder interviews, and reference to third-party sources. This included an assessment of national and regional energy policies, analysis of historical cookstove initiatives in India, and verification that no parallel mandatory programs exist in the project area.

Based on the above validation activities, the VVB concludes that the project satisfies the requirements for additionality under the applied methodology. The project clearly meets the regulatory surplus and positive list criteria of the Activity Method and is not driven by any legal or regulatory mandate within the host country.

3.3 Methodology Deviations

The project activity has not taken any deviations from the applied methodology VM0050 v1.0. Requantification is done in full compliance with the methodology. Also, no previously validated deviations exist.

3.4 Project Description Deviations

No project description deviations have been applied during the monitoring period selected for requantification.

3.5 Baseline and Monitoring

3.5.1 Baseline Reassessment

Is the project subject to a baseline reassessment?

☒ Yes

☐ No

The project has undergone a baseline reassessment following the transition from methodology VM0006 v1.2 to VM0050 v1.0 /B02/. In accordance with Section 2.3.2(2) of the VCS Methodology Change and Requantification Procedure v4.0 /B01/, the baseline scenario and baseline emissions have been reassessed to reflect the revised requirements of the applied methodology.

The VVB reviewed the updated Project Description (PD v1.8) /01/ and Requantification Report (v1.2) /03/ to evaluate the appropriateness, accuracy, and completeness of the revised baseline scenario in line with the VCS Program and VM0050 v1.0 /B02/. The VVB confirms that the application of VM0050 v1.0 /B02/ is appropriate for the project activity and that the rationale for the methodological transition is clearly presented.

The baseline reassessment has been conducted in accordance with Section 6.1 of VM0050 v1.0 /B02/. The updated PD /01/, particularly Section 3.4.1, provides a clear and well-substantiated justification for the continued validity of the baseline scenario. Although the project follows a common design framework, baseline surveys are conducted independently for each project instance to reflect local socio-economic and cooking conditions. For Instance 1, baseline data were collected across 11 mandals in the Alluri Sitarama Raju district of Andhra Pradesh between 3 November and 12-December-2022, prior to the distribution of improved cookstoves.

Baseline Survey Design and Sampling

To establish a robust sampling frame, a pre-baseline assessment was conducted to estimate the number of potential ICS beneficiary households in each village. Based on this assessment, random sampling was applied to select 238 villages for baseline assessment, within which 5 -10 households per village were surveyed. This approach resulted in deliberate oversampling to enhance statistical robustness, with a total of 1,944 households surveyed /23/, representing approximately 4% of all households in the project instance. This significantly exceeds the minimum sample size of 36 households /32/ required to achieve a 90/10 confidence and precision level.

Baseline surveys were implemented by trained local field teams using digital data collection tools. Survey records were signed, uploaded to a cloud-based database, and subjected to data validation and cleaning procedures prior to finalisation. The final baseline dataset was submitted to the VVB and Verra for validation. The VVB confirms that the sampling approach, survey implementation, and data management procedures are consistent with the requirements of VM0050 v1.0 /B02/ and applicable CDM best-practice guidance for household energy surveys /B04/.

Updated Data Collection under VM0050

VM0050 v1.0 /B02/ introduces a binding baseline questionnaire template that includes two additional mandatory questions related to household cooking frequency and variability, namely:

- (i) the number of meals prepared on the previous day and whether this is representative of typical cooking practices; and
- (ii) differences in the number of meals prepared on a typical weekend day.

The VVB confirmed that these questions were not included in the original baseline surveys conducted under VM0006 v1.2 /B06/. To ensure full compliance with VM0050 v1.0 /B02/ requirements, the Project Proponent conducted an additional 151 baseline household surveys together with 151 Kitchen Performance Tests (KPTs) /22/ between 16 November and 13 December 2024 and between 14 and 22 June 2025. These surveys explicitly incorporated the newly required questionnaire elements.

The VVB reviewed the updated survey instruments and verified that both questions were consistently applied across the additional sample. The results were incorporated into the updated baseline dataset and reflected in Section 3.4.2 of the revised PD /01/. During the onsite audit, the VVB applied acceptance sampling, as described in Section 2.4 of this report, to verify the implementation of the additional surveys and KPTs and confirmed that no discrepancies were identified. The combined dataset provides comprehensive, methodologically consistent, and VM0050-compliant baseline information for Instance 1.

Control Households

Of the 151 households included in baseline KPTs, 51 households were designated as control households and excluded from ICS distribution. These households serve as a reference group for monitoring external trends in fuel consumption and cooking behaviour, thereby strengthening the robustness of the baseline scenario and supporting long-term comparability in line with Section 6.2 of VM0050/B02/.

Baseline Conditions

The baseline survey database /23/ indicates that households in the project area rely exclusively on firewood for cooking, collected directly from nearby forests. LPG and other clean fuel users were not included in the project boundary. Average stove usage is approximately 4.4 hours per day, primarily indoors, with cooking and fuel collection activities predominantly undertaken by women. Respondents reported significant health, safety, and livelihood burdens associated with traditional stove use, including exposure to indoor air pollution, long fuel collection distances (average 3.31 km), physical strain, and wildlife-related risks. Over 98% of surveyed households expressed dissatisfaction with their existing stoves and willingness to adopt an improved cookstove.

The VVB confirms that these findings provide a credible, representative, and contextually accurate depiction of baseline conditions in the project area.

Baseline Fuel Consumption and Emissions

Unlike the previously applied methodology, VM0050 v1.0 /B02/ requires explicit quantification of baseline fuel consumption and baseline emissions. Section 4.1 of the PD provides this quantification. The VVB reviewed the underlying calculations, assumptions, and parameters and confirms that they are transparent, conservative, and consistent with VM0050 v1.0 and VCS Program requirements /B01/.

Baseline fuelwood consumption was determined to be 13.22 kg/household/day (equivalent to 4.83 t/year), based on KPT results. In accordance with Clarification 1 of the Corrections and Clarifications to VM0050 Energy Efficiency and Fuel-switch Measures in Cookstoves, v1.0 /B01/, baseline fuel consumption results were compared with data from reputable and recent external sources collected within the last five years. An independent third-party study conducted in the project area between December 2024 and January 2025 by Koshish Sustainable Solutions, a specialist local organisation with demonstrated expertise in biomass cookstove assessments, reported a comparable average baseline fuelwood consumption of 14.5 kg/household/day. The close alignment of these values demonstrates consistency with the clarification requirements and supports the credibility and representativeness of the project's baseline fuel consumption estimates.

The VVB verified that no new relevant national or sectoral policies have come into force since the initial baseline survey that would mandate a transition away from traditional biomass cooking technologies. The baseline scenario therefore remains consistent with the prevailing regulatory framework.

Based on the above assessment, the VVB concludes that the baseline reassessment has been conducted in full accordance with VM0050 v1.0, the VCS Methodology Change and Requantification Procedure, and the VCS Program requirements. The updated baseline scenario is considered realistic, representative, conservative, and supported by credible and recent data. The explicit inclusion of newly required baseline questionnaire elements and the designation of control households further strengthen the integrity and long-term validity of the project's emission reduction claims.

3.5.2 Baseline Scenario

Is the project applying a new methodology where the baseline scenario determination requirements are different from original methodology?

☒ Yes ☐ No

The project applies VM0050 v1.0 /B02/, which introduces revised baseline scenario determination requirements compared to VMR0006 v1.2. In accordance with Section 6.1 of VM0050 v1.0 /B02/, the baseline scenario was reassessed using the prescribed three-step approach to identify the most realistic and conservative scenario that would occur in the absence of the project activity.

Step 1: Identification of Alternative Baseline Scenarios

Three plausible alternatives were considered:

- **Alternative 1:** Implementation of the proposed ICS project without GHG revenue.
- **Alternative 2:** Introduction of LPG-fuelled cookstoves.
- **Alternative 3:** Continuation of traditional practices involving the use of woody biomass through inefficient stoves.

Step 2: Consideration of Policies and Regulations

The VVB assessed existing and planned national and regional policies and confirmed that no mandatory regulations require households in the project area to transition to improved cookstoves or cleaner fuels. Accordingly, none of the identified alternatives were excluded at this step.

Step 3: Assessment of Barriers

- **Alternative 1** was eliminated due to financial barriers, as the project's free distribution and servicing of ICS would not be viable without carbon finance.

- **Alternative 2** was eliminated due to institutional and information barriers, including difficulties in accessing LPG in remote project areas, high fuel costs, and a lack of user awareness and acceptance.

These barriers were assessed based on real-world implementation conditions and aligned with the criteria outlined in Section 6.1 of VM0050 /B02/ and VT0008 /B03/.

The sole remaining and selected scenario—Alternative 3: the continued use of traditional biomass cookstoves—is considered both conservative and credible. This scenario reflects the prevalent cooking practices in the project regions, which typically involve three-stone stoves or fixed clay stoves fuelled by firewood.

The revised baseline scenario is supported by comprehensive baseline surveys conducted across 11 mandals between November and December 2022, supplemented by additional household surveys and KPTs conducted between November 2024 and June 2025 to comply with updated VM0050 requirements. The inclusion of control households further enhances the robustness and credibility of the baseline scenario.

The VVB concludes that the baseline scenario has been identified and justified in accordance with VM0050 v1.0 and the VCS Standard. The selected scenario is realistic, conservative, and representative of conditions in the project area. The revised baseline scenario determination requirements have been fully met, and no material deficiencies were identified that would compromise the validity of the project's emission reduction claims.

3.5.3 Monitoring Plan

The project involves the large-scale distribution and sustained use of improved cookstoves (ICS) among rural households. A monitoring plan has been developed in compliance with the applied methodology VM0050 v1.0 /B01/, ensuring consistent, accurate quantification of emission reductions through data-driven performance tracking.

A statistically robust sampling strategy is adopted by the Project Proponent (PP), aligned with the CDM Guidelines and Standards on Sampling and Surveys of CDM Project Activities and Programmes of Activities (/ B04/). As a large-scale activity, the plan applies:

- **90/10 confidence/precision** for all proportion parameters
- **90/10 confidence/precision** for mean value parameters

To enhance methodological rigor, the PP applies a minimum 90/10 confidence/precision across all sampled parameters.

A **Simple Random Sampling (SRS)** method is applied for parameter monitoring, using data from the ICS registry as the sampling frame. Sample sizes are calculated to meet required confidence/precision thresholds, and household visits are conducted for field verification.

Monitoring surveys include both qualitative and quantitative assessments. The following table summarizes the key parameters, descriptions, monitoring frequency, and methods applied:

Parameter	Description	Monitoring frequency	Monitoring methods to be applied
$n_{j,k,y}$	Proportion of commissioned project devices of type j from batch k that are still being used regularly in year y	Expected annually	Site visits with visual verification and structured end-user surveys on ICS usage and condition
$BC_{b,i,y}$	Average quantity of fuel per baseline device type i during year y	Expected annually and will occur at least biennially	In-situ Kitchen Performance Tests (KPTs) /22/ over four days; festivals/holidays excluded; results must meet 90/10 confidence/precision
$BC_{p,j,k,y}$	Average quantity of fuel used by project device type j from batch k during year y	Expected annually and will occur at least biennially	KPTs /22/ conducted post-distribution for project households using ICS under similar protocol to baseline KPTs
$Hh_i / Hh_{j,k}$	The average household size of the target population using baseline device type i and project device type j from batch k .	A new value is determined every time the project activity is undertaken in a new instance	Household demographic data collected during KPTs; adult male equivalent calculated per Kitchen Performance Test Protocol v4.0.
$\eta_{new,avg,y}$	Weighted average efficiency of project devices in year y	Expected Annually	WBts per Protocol v4.2.3 on 3 ICS units $\times$ 3 replicates (9 tests) using simple random sampling from the first batch. Student's t-distribution applied (<30 samples) to meet 90/10 confidence/precision; additional tests if needed. For $y=0$, manufacturer-certified values used per VM0050 v1.0 (Option 2).

The monitoring plan is consistent with the requirements of methodology VM0050 v1.0 /B02/ and associated tools. The combination of proprietary digital tools (Tramontana ICS app), rigorous field-based monitoring, and a statistically sound sampling framework ensures comprehensive, reliable, and conservative data collection. The plan supports accurate emission reduction estimations and provides a credible basis for verification throughout the project lifecycle.

4 VERIFICATION FINDINGS

4.1 Project Details

Item	Evidence gathering activities	evidence checked	assessment conclusion
Audit history	Joint validation and verification of MP1 (21-January-2023 – 23-August-2023)	- VCS JPDMR v1.5 - Joint validation and verification report v3 - VCS Issuance deed of representation - VCS verification deed of representation	Based on the evidence checked along with other supportive documents (such as MP1 distribution database, baseline and project survey records, stakeholder consultation reports , etc.) the audit history has been confirmed.
Project Type	Reviewed Project Description (PD), assessed the technology type and applied methodology	Project Description (v1.1); Methodology VM0050 v1.0; ICS technical specification sheets	The project is correctly categorized as a large-scale GHG mitigation activity involving the distribution and monitoring of improved cookstoves. This classification aligns with the applied methodology and VCS definitions.

Project Proponent (PP) Identity	Reviewed corporate registration /14/, organizational structure /29/, and declarations	Company registration certificates /14/, signed declarations, legal representation documents	Based on the review of the above evidence, the Project Proponent (PP) has been verified as a legally registered entity /14/ with proper governance structure /29/ and authorized signatories. All submitted documents are authentic and satisfy the verification requirements. No discrepancies or inconsistencies were observed. The PP's identity is therefore confirmed and validated.
Project Start Date	Reviewed installation records, the Tramontana ICS database, signed user agreements, distribution records, and the metadata of photos from the first recorded entry /04/.	Tramontana ICS database entries, signed user agreements, distribution records, and metadata of photos from the first recorded entry /04/.	The project start date is confirmed as 21st January 2023, aligned with the first ICS distribution and verified through database records and photo metadata /04/. All evidence is consistent, authentic, and meets VCS Standard requirements.
Project Crediting Period	Assessed against VCS Program rules and methodology guidance	Crediting period outlined in PD; methodology timeline eligibility	The proposed crediting period, from 21-January-2023 to 20-January-2030, conforms to VM0050 v1.0 and VCS Program Rules (e.g., renewable or fixed period). The duration is clearly justified.

Project Boundary	Reviewed PD maps, KML file GPS data /27/, and ICS household coordinates.	KML file /27/ and GIS data exports, field maps, Tramontana system household locations	The geographic and physical boundary is clearly delineated. All emission sources/sinks relevant to the project activity are correctly included.
Ownership and Right to Claim Reductions	Examined ICS license agreements and credit ownership terms /06/, the contract between the PP, manufacturer, and the agent (Crop Zone) representing the PP's interests /33/.	Signed license/consent agreements /06/ CRS-23 Purchase Order (Sep 2022) /33/	Ownership of emission reductions is clearly assigned to the PP. End-user agreements confirm voluntary participation and transfer of rights. The CRS-23 Purchase Order with the manufacturer and the agreement /32/ with Crop Zone further substantiate the PP's entitlement and representation, ensuring no conflicting claims over the emission reductions.
Stakeholder Consultation	Assessed documentation of community meetings, feedback forms, and engagement activities	Meeting minutes, stakeholder lists, feedback logs, photos, response follow-ups	Stakeholder consultation was appropriately conducted per VCS requirements. Community feedback was documented and considered /13/.
Environmental Impact Assessment	Reviewed environmental risk (if any) for the project context	An Environmental Impact Assessment (EIA) is not required as there are only positive environmental outcomes	No significant negative environmental impacts were identified. Compliance with applicable laws and regulations is confirmed.

Project Additionality	Evaluated additionality assessment described in PD	VM0050 VT0008	The project is additional by positive list as per the methodology and supported by regional data and baseline study, baseline and project device KPT data, sales receipt, etc..
Leakage Assessment	Reviewed leakage risks and mitigation measures identified in the PD /01/	PD analysis /01/, kitchen fuel use data, WBT/KPT comparisons, double usage risk control	Leakage is appropriately addressed and controlled. Survey methods and serial tracking reduce risks of double counting or ICS stacking.
GHG Emission Reductions Quantification	Cross-checked emissions calculation methodologies, formulae, parameters, and sources	Emission factors, stove efficiency results (WBTs), fuel usage data (KPTs), sampling reports	Emission reductions are quantified in accordance with VM0050 v1.0 /B02/. All formulas and inputs conform to the methodology.

4.2 Accuracy of Reduction and Removal Calculations

Data and Parameters monitored:

Parameter	Description	Monitoring frequency	Means of assessment by VVB
$N_{j,k,y}$	Number of commissioned project devices of type j from batch k in year y	At the time of commissioning/ distribution of ICS	The VVB verified the commissioning of project devices via the Tramontana ICS distribution database/09/, confirming the number and type of devices deployed. A total of 46,160 ICS units were distributed, as verified through distribution records and signed delivery receipts. The distribution records were cross-checked with the implementation plan to ensure alignment with reported batches.

Parameter	Description	Monitoring frequency	Means of assessment by VVB
$n_{j,k,y}$	Proportion of commissioned project devices of type j from batch k that are still being used regularly in year y	Expected annually	Based on the monitoring survey, the VVB confirmed active usage of ICS units through document review, household interviews, and direct observations /19/. Spot checks and field records verified that the units were operational, with positive user feedback and no significant breakdowns recorded in the grievance logs maintained by the PP. In accordance with Clarification 2 of VM0050 v1.0, the usage rate parameter ($n_{(j,k,y)}$) has been conservatively capped at 90% /B02/. The VVB further assessed the customer support actions implemented by the PP across the entire target population and confirms that the PD /01/ and Requantification Report /03/ provide adequate justification demonstrating compliance with all three mandatory customer support criteria. This includes: (i) the selection of the Tramontana ICS as an appropriate technology that fully meets local cooking needs and uses the same baseline fuel, thereby minimizing stove stacking risks; (ii) the provision of user training materials and operational guidance in the local language; and (iii) the establishment of local maintenance teams, free repair and replacement services, and accessible communication channels for end users. These measures are supported by relevant documentary evidence and field records /28/. Based on the above assessment, the VVB concludes that the application of the 90% usage rate cap is methodologically appropriate and consistent with the requirements of VM0050 v1.0 and its associated Clarifications.
n_{HH}	Number of ICS distributed to each household	At the time of commissioning/	The VVB reviewed the project's stove distribution database /09/ and

Parameter	Description	Monitoring frequency	Means of assessment by VVB
		distribution of ICS	confirmed that each household is uniquely identified and, as a standard practice, receives one ICS unit. Identity verification documents (e.g. Aadhaar and BPL ration cards) are used by the Project Proponent to ensure household uniqueness and avoid duplicate registrations. In the rare cases where more than one stove is distributed to a single household (e.g. due to household size), this is explicitly recorded in the database and transparently accounted for through the application of the parameter n_{HH} the emission reduction calculations. The VVB verified the correct application of this parameter during the requantification calculation review. The distribution count of 46,160 ICS units was confirmed to match the number of recipient households. Based on the above verification activities, the VVB concludes that adequate controls are in place to prevent overlaps and double claiming of emission reductions at the household level.
$f_{NRB,y}$	Fraction of woody biomass that is established to be non-renewable used by device in year y	A new value is determined every time the project activity instance is undertaken in a new state	The VVB reviewed state-specific NRB values based on peer-reviewed literature and/or government publications submitted in the revised PD report. In addition, the fNRB Excel sheet /20/ was assessed to validate consistency with referenced sources. For each new implementation region, the NRB factor was found to be justified and compliant with VM0050 v1.0 /B02/ and VCS Program rules /B01/.
$BC_{b,i,y}$	Average quantity of fuel per	Expected annually and	The VVB assessed baseline fuel consumption through structured

Parameter	Description	Monitoring frequency	Means of assessment by VVB
	baseline device type i during year y	will occur at least biennially	monitoring surveys. KPT records /22/ were sampled and analyzed to calculate average fuel use for traditional stoves. These findings were cross-verified with kitchen observations and user recall data collected during field verification activities.
$BC_{p,j,k,y}$	Average quantity of fuel used by project device type j from batch k during year y	Expected annually and will occur at least biennially	The VVB confirmed project fuel consumption by evaluating monitored data sets and KPT results collected from sampled ICS users. Stove usage logs, user interviews, and observational data were reviewed to ensure methodological consistency with VM0050 v1.0 and the approved monitoring plan.
$Hh_i / Hh_{j,k}$	The average household size of the target population using baseline device type i and project device type j from batch k .	A new value is determined every time the project activity is undertaken in a new instance	Household size data was obtained through statistically representative surveys and validated against regional census benchmarks. Sampling protocols and calculations were reviewed and found to be in alignment with methodological specifications under VM0050 v1.0.
$\eta_{new,avg,y}$	Weighted average efficiency of project devices in year y	Expected Annually	The VVB reviewed efficiency test reports conducted in accordance with the Water Boiling Test (WBT) protocol. The weighted average thermal efficiency was confirmed from manufacturer specifications/18/ and independent laboratory results/24/. The findings were consistent with the required performance criteria under VM0050 v1.0.

The VVB confirms that the GHG emission reductions and removals reported in the project's GHG statement have been accurately quantified in accordance with the applied methodology VM0050 v1.0 and the monitoring plan. All parameters were monitored using verifiable methods, default values conformed to VCS Program rules, and calculations were found to be consistent, accurate, and free from transcription errors.

4.3 Quality of Evidence to Determine Reductions and Removals

The assessment team confirms the adequacy and reliability of the evidence used to determine GHG emission reductions and carbon dioxide removals under the revised methodology VM0050 v1.0. The quality and quantity of data reviewed from the new PD /01/ and the Requantification Report demonstrate transparency, consistency, and traceability, enabling a robust verification process.

During the field and desk assessments, the VVB evaluated the complete information flow—from data collection at the household level, through aggregation and calculation, to final inclusion in the monitoring report. The process was found to be systematic and coherent with the monitoring plan under the applied methodology. Key datasets, including the ICS distribution database, user survey responses, efficiency testing reports, and fuel use monitoring data, were reviewed and cross-validated for consistency and reliability.

Sources of evidence included documented records, field-level data, photographic evidence, grievance logs, and signed beneficiary forms. These were primarily internal but supported by external, third-party efficiency test results and referenced literature for fNRB estimation. All sources were found to be appropriate and reliable for verification purposes.

In cases where the methodology did not explicitly define calibration frequency for monitoring instruments, the calibration approach adopted by the project was assessed against good practice and found to be appropriate.

The VVB concludes that the evidence presented in the revised PD /01/ and Requantification Report /03/ is sufficient in quantity and appropriate in quality to support the reported emission reductions and removals. The evidence demonstrates adherence to the revised monitoring plan and methodology requirements under VM0050 v1.0 /B02/, ensuring a high level of confidence in the reported outcomes.

4.4 Non-Permanence Risk Analysis

This is not applicable to the project activity as the Project is not an AFOLU (Agriculture, Forestry and Other Land Use) project.

4.5 Requantification Validation and Verification Summary

Carbon Check conducted the methodology change and requantification verification of the emission reductions for the VCS project activity titled “Tramontana ICS-01 – The Distribution of Improved Cookstoves to Indigenous Communities across Rural India” (VCS ID – 4457) for the monitoring period 21-January-2023 to 23-August-2023, in accordance with the requirements under VCS Standard Version 4.7 /B01/. This verification was necessitated by the project’s transition from VMR0006 v1.2 /B06/ to the new methodology VM0050 v1.0 /B02/, prompting revisions to the baseline scenario, emissions quantification approach, and overall monitoring framework.

The **Project Proponent (PP)**, **Tramontana Impact Limited**, is responsible for the implementation of the project, the collection and management of monitoring data, and the accurate reporting of greenhouse gas (GHG) emission reductions, in accordance with the revised monitoring plan.

Carbon Check, as an accredited Validation and Verification Body (VVB) under a body recognized by the **International Accreditation Forum (IAF)**, has undertaken the independent validation and verification of the GHG statement with a **reasonable level of assurance**, in accordance with the provisions of **ISO 14064-3:2019**.

During the verification process, Carbon Check assessed the revised project documentation and confirmed its alignment with the eligibility and applicability conditions of **VM0050 v1.0**, alongside the applied tools:

- **CDM AMS-II.G v13.1** – Energy efficiency measures in thermal applications of non-renewable biomass /B02/
- **VT0008 v1.0** – Additionality Assessment /B03/
- **CDM TOOL30 v4.0** – Calculation of the fraction of non-renewable biomass /B03/

These methodologies and tools were reviewed for consistency with the revised Project Description and the characteristics of the project activity. The assessment confirmed that the methodological requirements, including scope, technology type, and emission sources, are appropriately met.

These methodologies and tools were thoroughly reviewed for consistency with the revised Project Description and the Requantification Report. The assessment confirmed that the methodological requirements—including applicability conditions, technological scope, emission sources, and quantification procedures—are fully satisfied and appropriately aligned with the project’s design and implementation.

The project's **updated baseline scenario**—the continued use of **non-renewable woody biomass in traditional cookstoves**—was validated through comprehensive field survey data and substantiated using third party kitchen performance test (KPT) study report data /24/. The VVB confirms that all steps undertaken by the PP align with the multi-step baseline identification and justification process outlined in Section 6.1 of VM0050 v1.0 /B02/.

To meet the **revised data and indicator requirements of VM0050 v1.0**, the Project Proponent conducted new baseline surveys, addressing previously unreported parameters. These efforts significantly enhanced the quality of baseline data. Furthermore, the project has adopted proprietary **mobile applications and digital systems** for data collection and verification, which have improved **traceability, transparency, and data accuracy** across all stages of monitoring.

Notably, unlike VMR0006, which assumed zero baseline emissions, VM0050 v1.0 requires quantification of baseline emissions. The project now applies an evidence-based approach, calculating baseline emissions for each household individually. These calculations have been reviewed and found to be accurate and conservative.

Based on the assessment conducted, Carbon Check confirms that:

- The project activity conforms fully with the requirements of VCS Version 4.7 and the applied methodology VM0050 v1.0, as well as all associated tools and guidance.
- The implementation is consistent with the revised and validated Project Description, with no material discrepancies observed.
- The monitoring plan, data collection procedures, and digital verification systems provide a high degree of transparency, integrity, and traceability.
- The updated emission reductions reported for the monitoring period are credible, conservative, and free from material misstatements.

Based on the validation and verification activities undertaken, Carbon Check (India) Private Limited concludes that the project activity conforms to all validation and verification criteria set under VCS Version 4.7, and that the GHG statement is a fair and accurate representation of the emission reductions achieved during the monitoring period. No qualifications or adverse findings were identified. The project is deemed to have been implemented in accordance with the approved methodology and revised Project Description.

4.6 Validation Conclusion

The data and information supporting the GHG statement assertion are a combination of historical and projected data. Historical data include the field survey results, fuel usage patterns, stove types, and socioeconomic indicators collected through structured baseline surveys and validated using region-specific literature, credible government statistics, and site verification. Projected data primarily relate to the anticipated emission reductions based on expected stove performance, usage patterns, and the continued displacement of non-renewable biomass through the adoption of improved cookstoves (ICS).

The assumptions and methods used to estimate the future GHG emission reductions are considered reasonable and are consistent with the requirements of the applied methodologies:

- VM0050 v1.0 – Energy Efficiency and Fuel-Switch Measures in Cookstoves,
- CDM AMS-II.G v13.1 – Energy Efficiency Measures in Thermal Applications of Non-Renewable Biomass,
- VT0008 v1.0 – Additionality Assessment Tool, and
- CDM TOOL30 v4.0 – Calculation of the Fraction of Non-Renewable Biomass.

These methodologies and tools were reviewed by Carbon Check (India) Private Limited (Carbon Check) for consistency with the revised Project Description and the characteristics of the project activity and requantification report. The assessment confirmed that the methodological requirements, including scope, technology type, and emission sources, are appropriately met.

The assumptions include, but are not limited to, projected stove usage rates, continued fuelwood access and consumption patterns, and stove performance over time. Carbon Check confirmed that these assumptions are grounded in field realities and supported by conservative estimates, robust data collection methods, and standardized monitoring procedures. Nonetheless, it is recognized that actual results may vary due to uncertainties such as behavioral changes in stove usage, regional developments affecting biomass availability, or unforeseen socioeconomic shifts.

Based on the validation activities performed, Carbon Check concludes that the project is likely to achieve the estimated GHG emission reductions as described in the updated Project Description. No material discrepancies or inconsistencies were identified during the validation process.

The GHG reductions claimed are based solely on emission reductions (and not removals) for this activity, and have been validated in full accordance with the requirements under VCS Version 4.7.

Crediting Period: From 21-January-2023 to 20-January-2030.

Validated estimated GHG emission reductions and carbon dioxide removals for the project crediting period:

Vintage period	Estimated baseline emissions (tCO ₂ e)	Estimated project emissions (tCO ₂ e)	Estimated leakage emissions (tCO ₂ e)	Estimated reduction VCU (tCO ₂ e)	Estimated removal VCU (tCO ₂ e)	Estimated total VCUs (tCO ₂ e)
21-January-2023 to 31-December-2023	201,074	85,628	5,772	109,674	—	109,674
01-January-2024 to 31-December-2024	852,303	332,512	25,990	493,801	—	493,801
01-January-2025 to 31-December-2025	1,552,235	586,448	48,289	917,498	—	917,498
01-January-2026 to 31-December-2026	1,802,708	707,070	54,782	1,040,856	—	1,040,856
01-January-2027 to 31-December-2027	1,803,404	761,783	52,081	989,540	—	989,540
01-January-2028 to 31-December-2028	1,803,404	826,284	48,856	928,264	—	928,264
01-January-2029 to 31-December-2029	1,803,404	904,068	44,967	854,369	—	854,369
01-January-2030 to 20-January-2030	100,189	50,468	2,486	47,235	—	47,235
Total	9,918,722	4,254,261	283,223	5,381,237	—	5,381,237

4.7 Verification Conclusion

Carbon Check has conducted the verification of the quantity of GHG emission reductions achieved by the project activity during the requantification verification period, as detailed in the Requantification Report. The level of assurance provided is reasonable assurance, consistent with the requirements of VCS Version 4.7 and ISO 14064-3:2019.

The verification covers the verification period from 21-January-2023 to 23-August-2023. Carbon Check confirms that the reported GHG emission reductions, expressed in tonnes of CO₂ equivalent (tCO₂e), for the verification period are free from material misstatement and fairly represent the reductions achieved by the project in line with the approved methodology VM0050 v1.0 and the validated Project Description.

During this verification period, the project reports only GHG emission reductions and no removals. No material discrepancies or inconsistencies were identified throughout the verification process.

Carbon Check concludes that the GHG statement for this verification period is a fair and accurate representation of the emission reductions achieved by the project.

Verification periods being requantified: From 21-January-2023 to 20-January-2030.

Verified GHG emission reductions and carbon dioxide removals in the above verification period(s):

Verification Period	21-Jan-2023 to 23-Aug-2023					
Vintage period	Baseline emissions (tCO ₂ e)	Project emissions (tCO ₂ e)	Leakage emissions (tCO ₂ e)	Reduction VCU (tCO ₂ e)	Removal VCU (tCO ₂ e)	Total VCUs (tCO ₂ e)
21-January-2023 to 31-August - 2023	52,560	21,894	1,533	29,133	0	29,133
Total	52,560	21,894	1,533	29,133	0	29,133

4.8 Ex-ante vs Ex-post ERR Comparison

Vintage period	Ex-ante estimated reductions/removals	Achieved reductions/removals	Percent difference	Explanation for the difference
21-January-2023 to 23-August-2023	42,662	29,133	−31.7%	The VVB has reviewed the reported emission reductions for the vintage period 21-January-2023 to 23-August-2023 under Instance 1 of the project activity. The ex-ante emission reductions were estimated at 42.662 tCO₂e, while the actual ex-post reductions achieved during the 1st monitoring period were reported as 29,133 tCO₂e, reflecting a 31.7% variance. The deviation is primarily attributed to the following verified and justified factors: • Application of a lower energy savings cap $EC_{i,y}$ of 0.0656 TJ in ex-post calculations, based on field measurements through Kitchen Performance Tests (KPTs), as required under the applied methodology. The ex-ante estimate had used a theoretical value of 0.0753 TJ, leading to a conservative revision of energy savings. • Delays in stove distribution caused by heavy monsoon conditions between June and August 2023, which impeded access to remote project areas. • Resurvey requirements due to identified data quality issues in initial household records, resulting in slower-than-expected progress. The corrective measures applied by the Project Proponent (PP) — including retraining of field teams and stringent data validation protocols — are found to be in line with good monitoring practice and contribute to data reliability.

				The VVB confirms that all emission reductions were quantified in accordance with the applied methodology (VM0050 v1.0), using credible data sources and field-verified measurements. The monitoring plan was adhered to, and the emission reductions have been appropriately adjusted based on actual stove usage, adoption rates, and field conditions. Based on the documentation review, on-site assessments, and audit of the Monitoring Report, the VVB concludes that the reported emission reductions of 29,133 tCO₂e for the monitoring period 21-January to 23-August-2023 are: • Conservatively estimated in line with methodological requirements; • Supported by reliable field data, validated through multiple quality assurance layers; • Reflective of actual project performance, considering on-the-ground challenges and operational delays; • Eligible for issuance under the applied methodology and VCS Program rules. The observed variance from ex-ante projections is fully explained, justified, and accepted by the VVB. The VVB finds no material discrepancies or non-conformities that would compromise the integrity of the emission reduction claim for the first monitoring period of Instance 1.
Total	42,662	29,133	31.7%	

4.9 VCU Reconciliation Summary Table

Vintage period	Approved total VCUs eligible for issuance	New total VCUs eligible for issuance	Reconciliation percentage (%) = New VCUs / original VCUs	Original total buffer pool allocation	New total buffer pool allocation	Buffer allocation difference (new total – old total)
21-January-2023 to 31-August - 2023	62,743	29,133	46.4%	NA	NA	NA

APPENDIX 1: COMMERCIALLY SENSITIVE INFORMATION

No Commercially sensitive information provided by the PP during this monitoring period.

APPENDIX 2.1: REFERENCE DOCUMENTS

Ref	Document
/01/	VCS PD for the project “TRAMONTANA ICS-01 – The distribution of improved cookstoves to indigenous communities across rural India” version 1.8, dated 02-January-2026
/02/	a) ER calculation spread sheet corresponding to /01/ “ICS-01 Ex-Ante vs Achieved Calculations M2”. b) ER calculation spread sheet corresponding to /03/ “ICS-01 Requantification Calculations”.
/03/	VCS Requantification Report, version 1.2, dated 02-January-2026
/04/	Registration cum consent deed as evidence for the start date of the project.
/05/	Registration cum consent deed as evidence for unique identification of each of the ICS.
/06/	Proof of right of VERs by (license agreement between PP and end-user)
/07/	Monitoring survey questionnaire template.
/08/	Monitoring survey records for the monitoring period. • ICS-01 Monitoring Database MP1
/09/	Database for the ICS distribution records for the monitoring period.
/10/	Sample size calculation and precision level achieved calculator for the monitoring period.
/11/	Random sample selection for the monitoring survey parameters.
/12/	Training records; 11-November-2022 • Attendance register • Training Images
/13/	Stakeholder Consultation MP1 • Meeting Minutes and Attendance sheets
/14/	Company registration certificate • Tramontana Impact Registration • CropZone Certificate of Incorporation
/15/	PPs declaration that; • Project is not part of any other GHG program. • Project will not claim emission reductions under any other program. • No project device will be double counted. • PP has right of verified emission reductions.

/16/	Technical specifications of the ICS including the life span. - CRS-23 model ICS - STAR-25 model ICS
/17/	Evidence of employment generated during MP1.
/18/	Manufacturer's brochure
/19/	Onsite visit documents - Attendance sheets - Assessment notes - Site visit Pictures
/20/	Evidence for fNRB calculation fNRB calculation_instance 1
/21/	Water Boiling Tests (WBT) Reports conducted by " <i>Thermodynamics Laboratory at the Department of Mechanical Engineering, Jorhat Engineering College (JEC)</i> " dated 17-January-2020.
/22/	Kitchen performance test (KPT) reports (excel sheet) a) ICS-01 KPT Instance 1
/23/	Baseline database ICS-01 Baseline Database Instance 1
/24/	Third Party Kitchen Performance Test (KPT) Study Report
/25/	Calibration of KPT tools - Digital Scales, Atom Aliston - Moisture meter
/26/	End-user license agreements
/27/	KML file for project boundary
/28/	Clarification 2 VM0050 supporting docs
/29/	Tramontana Asset Management Organizational Chart
/30/	Cropezone Human Resource Policy Manual
/31/	Contract between VVB and PP (Dated-04-October-2024)
/32/	Sample size calculation and precision level achieved calculator for the Baseline
/33/	Sale & Purchase Agreements: Tramontana Asset Management Limited, Ravi Engineering & Chemical Works, and CropZone Agro Forestry Private Limited

APPENDIX 2.2 : BACKGROUND DOCUMENTS

Ref	Document
/B01/	VCS Requirements - VCS Standard (v4.7, dated 16-April-2024) - VCS Program Guide (v4.4, dated 29-August-2023) - VCS Validation and Verification Manual version (v3.2, dated 19-October-2016) - Registration & Issuance Process (v4.5, dated 16-April-2024) - VCS Program Definitions version (v4.5, dated 16-April-2024) - VCS Methodology Change and Requantification Procedure v4.0, dated 10-October-2024 - VCS MR template version 4.4 - Correction and Clarifications To VM0050 Energy Efficiency and Fuel-switch Measures In Cookstoves, (v1.0 dated 03-February-2025)
/B02/	Applied baseline and monitoring methodology. VM0050 Version 1.0 “<i>Energy Efficiency and Fuel Switch Measures in Cookstoves</i>” - CDM AMS-II G Version 13.1 “<i>Energy efficiency measures in thermal applications of non-renewable biomass</i>”
/B03/	Methodological Tool - CDM Tool 30 “Calculation of the fraction of non-renewable biomass” Version 4.0 - VCS Tool: VT0008 “Additionality Assessment” Version 1.0
/B04/	“Standard for sampling and surveys for CDM project activities and programme of activities” (version 09.0) - Guidelines for sampling and surveys for CDM project activities and Programme of Activities (version 04)
/B05/	Website and links: - IPCC (http://www.ipcc-nggip.iges.or.jp) http://cdm.unfccc.int http://www.v-c-s.org
/B06/	Previous baseline and monitoring methodology. VMR0006. version 1.2 “Methodology for Installation of High Efficiency Firewood Cookstoves”

APPENDIX 3 : ABBREVIATIONS

CDM	Clean Development Mechanism
BE	Baseline Emission
CAR	Corrective Action Request
CC IPL	Carbon Check (India) Private Ltd.
CDM	Clean Development Mechanism
CL	Clarification Request
CO₂	Carbon Dioxide
CO_{2e}	Carbon Dioxide Equivalent
DVR	Draft Verification Report
EF	Emission Factor
ER	Emission Reduction
GHG	Greenhouse gas(es)
ICS	Improved Cook Stove
IPCC	Intergovernmental Panel on Climate Change
KPT	Kitchen Performance Test
NA	Not Applicable
OSV	On Site Visit
PD	Project Description
PP	Project Proponent
QC/QA	Quality control/Quality assurance
TR	Technical Review
UNFCCC	United Nations Framework Convention on Climate Change
VCS	Verified Carbon Standard
VCU	Verified Carbon Unit
VB	Validation Verification Body
VM	Validation and Verification Manual
WBT	Water Boiling test

APPENDIX 4 : CERTIFICATES OF COMPETENCE

Carbon Check (India) Private Limited

Certificate of Competency

Manas Halder

has been qualified as per CCIPL's internal qualification procedures in accordance with the requirements of CDM AS, A 6.4 AS/ ISO/IEC14065:2020, ISO/IEC 17029:2019 and other applicable GHG programs:
for the following functions and requirements:

- | | | | |
|--|---|--|---|
| ☐ Management Function | ☐ Administrative | ☒ Validator/Verifier | ☒ Team Leader |
| ☐ Acting Team Leader | ☒ Technical Expert | ☐ Technical Reviewer | ☐ Trainee Assessor |
| ☒ Assessor | ☒ Regional Expert | ☐ Gender Expert | ☐ Financial Expert |
| Bangladesh, I | | | |
| ☐ Plastic Waste Expert | ☐ Health Expert | ☐ CCB Expert | ☒ SDG Expert |
| ☒ Expert Social Aspect | ☒ Expert Environmental Aspect | ☐ Internal Auditor | ☐ Legal Expert |
| ☒ FOEN Approved Technical Expert | ☐ FOEN Approved Quality Officer | | |

in the following Technical Areas:

- ☐ PWRP 2 - Plastic Waste Collection & Recycling
- ☐ TA 1.1 - Thermal energy generation
- ☒ TA 1.2 - Renewable Energy generation
- ☐ TA 10.1 - Fugitive emissions from oil and gas
- ☒ TA 13.1 - Solid waste and Wastewater
- ☐ TA 13.2 - Manure
- ☐ TA 14.1 - Afforestation and reforestation (except REDD & REDD+)

- ☐ TA 14.1 - Afforestation and reforestation (only REDD & REDD+)
- ☐ TA 15.1 - Agriculture (With Model Validation)
- ☐ TA 15.1 - Agriculture (Without Model Validation)
- ☐ TA 16.1 Carbon Capture and Storage ☐ TA 2.1 - Energy Distribution
- ☒ TA 3.1 - Demand Side Energy Efficiency
- ☐ TA 4.1 - Cement and Lime Production
- ☐ TA 4.n - Waste Heat Recovery & Fuel Switch ☐ TA 5.1 - Chemical Industry
- ☐ TA 5.2 - Caprolactam, Nitric and Adipic Acid ☐ TA 7.1 - Transport
- ☐ TA 8.1 - Mining/Mineral Production
- ☐ TA 9.1 - Aluminium and magnesium production
- ☐ TA 9.2 - Iron, Steel and Ferro-alloy Production
- ☐ TA PWRP 1 - Plastic Waste Collection & Recycling

Issue Date 29-12-2025 **Expiry Date** 29-12-2026

Vikash Kumar Singh
Director - Compliance

Carbon Check (India) Private Limited

Certificate of Competency

Seshan Ranganathan

has been qualified as per CCIPL's internal qualification procedures in accordance with the requirements of CDM AS, A 6.4 AS/ ISO/IEC14065:2020, ISO/IEC 17029:2019 and other applicable GHG programs:

for the following functions and requirements:

- | | | | |
|---|--|--|---|
| ☐ Management Function | ☐ Administrative | ☒ Validator/Verifier | ☒ Team Leader |
| ☐ Acting Team Leader | ☒ Technical Expert | ☒ Technical Reviewer | ☐ Trainee Assessor |
| ☐ Assessor | ☒ Regional Expert | ☐ Gender Expert | ☐ Financial Expert |
| <div style="border: 1px solid black; padding: 2px 10px; display: inline-block;">India</div> | | | |
| ☐ Plastic Waste Expert | ☐ Health Expert | ☐ CCB Expert | ☐ SDG Expert |
| ☐ Expert Social Aspect | ☐ Expert Environmental Aspect | ☐ Internal Auditor | ☐ Legal Expert |
| ☐ FOEN Approved Technical Expert | ☐ FOEN Approved Quality Officer | | |

in the following Technical Areas:

- ☐ PWRP 2 - Plastic Waste Collection & Recycling
- ☒ TA 1.1 - Thermal energy generation
- ☒ TA 1.2 - Renewable Energy generation
- ☐ TA 10.1 - Fugitive emissions from oil and gas
- ☒ TA 13.1 - Solid waste and Wastewater
- ☒ TA 13.2 - Manure
- ☐ TA 14.1 - Afforestation and reforestation (except REDD & REDD+)

- ☐ TA 14.1 - Afforestation and reforestation (only REDD & REDD+)
- ☐ TA 15.1 - Agriculture (With Model Validation)
- ☐ TA 15.1 - Agriculture (Without Model Validation)
- ☐ TA 16.1 Carbon Capture and Storage ☐ TA 2.1 - Energy Distribution
- ☒ TA 3.1 - Demand Side Energy Efficiency
- ☐ TA 4.1 - Cement and Lime Production
- ☐ TA 4.n - Waste Heat Recovery & Fuel Switch ☒ TA 5.1 - Chemical Industry
- ☐ TA 5.2 - Caprolactam, Nitric and Adipic Acid ☐ TA 7.1 - Transport
- ☐ TA 8.1 - Mining/Mineral Production
- ☐ TA 9.1 - Aluminium and magnesium production
- ☐ TA 9.2 - Iron, Steel and Ferro-alloy Production
- ☐ TA PWRP 1 - Plastic Waste Collection & Recycling

Issue Date 12-12-2025 **Expiry Date** 12-12-2026

Vikash Kumar Singh
Director - Compliance

APPENDIX 5: FINDINGS LOG

Table 1. Remaining FAR from validation and/or previous verification

FAR ID	01	Section no.	Date: 10/07/2025
Description of FAR			
“Ex-ante Emission Reductions provided on the Verra page and in the PDMR are inconsistent. The estimated ERs are to be made consistent between the project documents and project activity web page. VVB to verify the same in the subsequent verification” With regard to the above FAR raised during joint validation and verification, the project proponent is requested to ensure that the updated version of PDMR uploaded to the Verra registry’s project page.			
Project proponent response			Date: 01/08/2025
The current ex-ante emission reductions number on the Verra page of 9,171,123 tCO₂e matches the latest PDMR submitted to Verra dated 1st August 2024. Upon submission of an updated PD to Verra, we will request the ex-ante emissions to be updated.			
Documentation provided by project proponent			
VVB assessment			Date: 07/08/2025
The VVB has reviewed the ex-ante emission reduction figure listed on the Verra Registry project page (9,171,123 tCO₂e over the first crediting period) and confirms that it matches the value stated in the most recent PDMR dated 16/07/2024. The PP has updated the PD due to the project’s transition from the previously applied methodology VMR0006 v1.2 to the updated VM0050 v1.0 and will request Verra to update the estimated ER on the project page upon submission of updated PD, and VVB to verify the same in the subsequent verification. FAR 01 from this verification is raised.			

FAR ID	02	Section no.	Date: 10/07/2025
Description of FAR			
“For the current project activity instance, the Alluri Seetharama Raju district has been covered in the baseline survey. A baseline survey has to be conducted before the implementation of any new PAIs in compliance with the applied methodology section 9.1.” With regard to the above FAR raised during joint validation and verification, the project participant is requested to ensure that a baseline survey is conducted before implementation of any new project activity instance and provide records of such survey performed and related supportive documents.			
Project proponent response			Date: 01/08/2025

In Section 3.4.2 of the PD, it has been explicitly stated that baseline surveys for Instance 1 were completed prior to distribution between 3rd November and 12th December 2022. The start date of distribution in Instance 1 was 21st January 2023, which can be verified from the database “ICS-01 Distribution Database M2”. This was used previously to register the project under the VMR0006 v1.2 methodology.

After the project start date, Verra released VM0050 v1.0 in October 2024. To transition from the earlier methodology (VMR0006 v1.2) to VM0050 v1.0, an additional 151 baseline surveys were conducted between 16th November and 13th December 2024, and again in June 2025. These surveys were necessary to comply with VM0050's additional requirements, which included two new questions (and Kitchen Performance Tests) which were not covered in the initial baseline surveys conducted.

Documentation provided by project proponent

VVB assessment

Date: 07/08/2025

The VVB reviewed the baseline survey documentation and confirms that the initial surveys were conducted between 3 November and 12 December 2022, prior to the start of distribution on 21 January 2023, in line with VMR0006 v1.2. Following the adoption of VM0050 v1.0, the PP conducted additional baseline surveys in November-December 2024 and June 2025 to address the updated methodological requirements, including two new survey questions and KPTs.

The VVB concludes that the baseline surveys were implemented appropriately and in compliance with Section 9.1 of VM0050, and that the FAR has been satisfactorily addressed with respect to the current verification.

In this regard, PP to comply with para 6.2 of VMR0006 v1.2 in conjunction with correction 1 of Corrections & Clarifications to VMR0006 v1.0 and must conduct a simple random sampling where a survey covers two or more project activity instances and the population is homogeneous, and where the population is not homogenous, stratified random sampling must be used. Also, follow-up baseline surveys must be conducted at most every five years from the date of the last survey in control households that do not participate in the project. Moreover, baseline survey performed prior to crediting period renewal must be conducted as physical on-site visit. VVB to verify this during subsequent verifications. FAR 02 from this verification is raised.

Table 2. CLs from this validation

CL ID	01	Section no.	1.18 PD	Date: 10/07/2025
Description of CL				
In section 1.18 of the PD, - For SDG 01 indicators 1.2.1 and 1.4.1, the explanation provided under the 'Contribution of Project Activity and Monitoring Approach' column does not sufficiently detail how the project intends to generate disaggregated data in alignment with the respective SDG indicator methodologies. The Project Proponent (PP) shall clarify how the monitoring approach will enable disaggregated reporting by sex and age for indicator 1.2.1, and how access to specific basic services, such as clean cooking, will be clearly defined, measured, and tracked in accordance with indicator 1.4.1. - The monitoring approach lacks specific health-related metrics aligned with SDG 3.9.1. PP shall define measurable indicators (e.g., reduced indoor air pollution exposure or respiratory symptoms), clarify the data collection methodology, and explain how this will demonstrate reduced health risks or mortality over time. - PP shall clarify how the monitoring approach intends to capture participation in formal and non-formal education and training disaggregated by sex for indicator 4.3.1, and how ICT skills will be defined, categorized, and tracked by type of skill in accordance with the methodology for indicator 4.4.1. - For SDG 05 indicator 5.4.1, PP shall clarify how the monitoring approach will capture disaggregated data by sex, age, and location on time spent on unpaid domestic and care work. Additionally, please elaborate on how the project will measure and report reductions in drudgery and negative impacts experienced by women and girls aged 15 years and older, including qualitative aspects such as well-being and safety during firewood collection. Details on the methodology to track female employment and training under the project should also be provided. - SDG 08 requires disaggregated data by sex, age, and disability status on employment under indicator 8.5.2. PP shall clarify how the monitoring approach captures this data. Additionally, please provide details on how training participation, job quality, and career development opportunities are measured and reported, including for both direct project staff and external service providers.				
Project proponent response				Date: 01/08/2025
For SDGs 1, 3, 4, 5 and 8, the SDG indicators have been replaced with project specific indicators which directly align with the monitoring approach described in section 1.18 of the PD. The update to the indicators ensures that the monitoring data collected can be used to directly measure the impact on the SDG. For SDGs 3 and 5, additional details have been added in Table 6 of the PD describing the monitoring approach used, clarifying the data collection methodology. Specific questions are asked during monitoring surveys which are used to assess the impact on the specific SDG indicator. For SDG 3, details have been added to the SDG table to: - Define the specific health-related metrics including how changes in air pollution and well-being are measured and, - Demonstrate how these measures are used to demonstrate reduced health risks For SDG 5, clarifications have been added to the SDG table to include: - How time savings from ICS use are monitored and, - How female employment is tracked, specifically with the use of employment records and contracts.				

Documentation provided by project proponent	
VCS4457 Project Description v1.7	
VVB assessment	Date: 07/08/2025
1. In the revised Section 1.18 of the Project Description (PD), the Project Proponent (PP) has removed SDG Indicator 1.2.1 and retained a focused contribution to SDG Target 1.4 through a project-specific indicator aligned with Indicator 1.4.1. The indicator—"Number of people provided access to the basic service of cleaner cooking through the provision of a Tramontana ICS"—is appropriate given that clean cooking is recognized as a core basic service under SDG 1.4.1. The monitoring approach, which includes identification of BPL households and tracking of ICS distribution and usage, is clearly described and relevant to the project scope. The revised indicator and its alignment with SDG 1.4 are acceptable. 2. The Project Proponent (PP) has revised its approach to SDG 3 by introducing a project-specific indicator aligned with SDG 3.9.1, measuring the proportion of households reporting reduced indoor smoke and general health improvements. While not directly measuring mortality or specific symptoms, the updated monitoring approach includes defined survey questions on air pollution and well-being, and tracks stove use in enclosed spaces to assess exposure. The revisions provide a reasonable proxy for reduced health risks from household air pollution and are acceptable for demonstrating contribution to SDG 3.9. 3. The updated Project Document (PD) has revised the SDG 4 indicators from the original SDG 4.3.1 (Participation rate of youth and adults in formal and non-formal education and training, by sex) and SDG 4.4.1 (Proportion of youth and adults with ICT skills, by type of skill) to project-specific indicators: a) Proportion of households reporting increased time spent on education due to project time savings (aligned with SDG target 4.3), and b) Number of adults receiving specific ICT training under the project (aligned with SDG target 4.4). These project-specific indicators and the monitoring approach are well-defined and align with the SDG targets. Based on this revision, the PP's response is acceptable. 4. PP has updated and defined 'Project Specific Indicator: Proportion of beneficiaries able to spend more time contributing to the main family income, spending time with family, and helping others in the community'. This aligns with SDG 5.4 which focuses on recognizing and valuing unpaid care and domestic work and acceptable. 5. The Project Proponent (PP) has introduced a project-specific indicator—"Number of people employed directly and indirectly as a result of the project"—to align with SDG Target 8.5. The updated Project Document provides sufficient detail on employment creation, training participation, and career development opportunities, including for both direct staff and external service providers. The monitoring approach is clearly described, and the response is acceptable.	

CL ID	02	Section no.	3.4.2 PD	Date: 10/07/2025
Description of CL				
In section 3.4.2 of the PD, it states that 1,944 households were surveyed but does not explain how the sample size was determined or whether it meets the required confidence and precision of 90/10. As per VM0050 and CDM AMS-II.G, surveys must follow the latest CDM Standard for Sampling and Surveys. PP shall clarify the sampling methodology, including statistical methods used, and correct any inconsistencies.				
Project proponent response				Date: 01/08/2025
The sample size was determined using the guidelines presented in Appendix 5 of VM0050 v1.0 and, in line with the CDM Standard for Sampling and Surveys, meets the 90/10 confidence/precision requirements. Section 5.3.2 has been updated to include the sample size calculation, methodology and confidence/precision analysis. Specifically, the minimum sample size for the baseline survey has been calculated in section 5.3.2.2 using the method described in Appendix 5 of VM0050 v1.0. In section 5.3.2.1, statistical analysis of a selected baseline parameter Hh_i has been conducted with the calculated precision demonstrating the 90/10 rule has been achieved. In addition to these updates, section 3.4.2 has been updated to include reference to section 5.3.2.				
Documentation provided by project proponent				
• Baseline Sample Size + Precision • Baseline Survey Instance 1 Random Sample				
VVB assessment				Date: 07/08/2025
The VVB has completed its assessment of the revised Project Description (PD) and the updated supporting spreadsheets: ICS-01 KPT Instance, Baseline Sample Size + Precision, and Baseline Survey Instance 1 Random Sample. The sampling methodology has been clarified and is now consistent with the requirements of VM0050 v1.0 and the CDM Standard for Sampling and Surveys. Section 5.3.2 of the PD provides a detailed explanation of the sample size determination, the statistical methodology applied, and confirmation that the 90/10 confidence/precision criterion has been achieved. In addition, Section 3.4.2 has been updated to reference the relevant methodological details. Based on the review, all requested clarifications have been adequately addressed. Accordingly, this CL is considered closed.				

CL ID	03	Section no.	4.4 PD	Date: 10/07/2025
Description of CL				
In section 4.4 of the PD, the explanation for calculating annual efficiency reduction is noted; however, the method used to determine the average efficiency values for each year remains unclear. The PP is requested to elaborate more on the calculation method.				
Project proponent response				Date: 01/08/2025

It should be noted that the ex-ante annual efficiency calculations are provided in detail for both stove models in the Excel sheet “ICS-01 Ex-Ante vs Achieved Calculations M2”. However, to improve clarity within the PD, two additional columns ($\eta_{new} Start$ and $\eta_{new} End$) have been added to the efficiency table in section 4.4. The two new columns are averaged to calculate the value of $\eta_{new,avg,y}$ presented in the final column. A description of the calculation has also been included in the paragraph above the table, with reference to the specific columns.

Documentation provided by project proponent

VCS4457 Project Description v1.7

VVB assessment

Date: 07/08/2025

The VVB has reviewed the revised PD v1.7 and supporting calculations. Section 4.4 has been updated to clarify the method for deriving average annual efficiency values using new columns ($\eta_{new} Start$ and $\eta_{new} End$), with an explanation added above the table. The clarification is satisfactory. This clarification is closed.

CL ID	04	Section no.	5.2 PD	Date: 10/07/2025
Description of CL				
In section 5.2 of the PD, - For parameter $n_{j,k,y}$, the PP has considered a 100% usage rate based on baseline survey results. However, according to the Corrections and Clarifications to VM0050 v1.0, Clarification 2: <i>“For projects employing surveys, the usage rate is capped at 90% where all of the following customer support actions are undertaken across the entire target population and demonstrated at verification:</i> <i>1) Selection of technologies and fuels that fully meet the cooking needs of the target population, demonstrated by citing robust research or conducting an investigation of cooking practices and attitudes</i> <i>2) Implementation of support activities to assist the target population in effectively operating and maintaining their cookstoves. These may include providing materials (print, in-person, or video) on how to operate the cookstove to prepare common local foods, how to troubleshoot common operational issues, and how to make minor repairs (including obtaining necessary replacement parts). All such communications and materials must be provided in local language(s) commonly used in the project area.</i> <i>3) Provision of a commonly used, toll-free communications channel through which the target population can contact the project proponent to access support (e.g., maintenance and repair services)</i> <i>For projects employing surveys and that do not implement all of these customer support actions across the entire target population, the usage rate is capped at 75%.”</i> PP shall demonstrate compliance with the above methodology requirement and rectify as required. - Under the QA/QC procedures of the relevant parameters, PP shall provide details on the internal quality checks (e.g., double entry verification, supervisor oversight, or re-surveys) that are implemented to ensure the accuracy of captured data. Please note, the same clarifications are also applicable to section 3.2 of the Requantification Report.				
Project proponent response				Date: 01/08/2025
- Section 5.2 and 5.3.2.1 of the PD and section 3.3.2 of the Requantification Report have been updated in line with the Correction and Clarifications to VM0050 v1.0 documentation to include a detailed description of how the project meets the $n_{j,k,y}$ cap requirements. Updates to parameter tables and emission calculations have also been completed to account for the change in the parameter value to 90%. - Updates to the “QA/QC procedures to be applied” field of the parameter tables in section 5.2 of the PD (and 3.2 of the Requantification Report) have been made to include details on the techniques and procedures applied to ensure accuracy of monitored parameters. Note the parameter $n_{j,k,y}$ <i>also includes further</i> details on field staff training, oversampling and the process of internal monitoring to verify data accuracy.				
Documentation provided by project proponent				

- ICS Instructions - Licence Agreement Example - Contact Details - Instance 1 Maintenance Team Roster - Spare Parts Folder - Household Visits Aug 2024 - VCS4457 Project Description v1.7	
VVB assessment	Date: 07/08/2025
- The VVB has reviewed the revised PD (Sections 5.2 and 5.3.2.1) v1.7 and revised Requantification Report (Section 3.3.2). The usage rate ($n_{(j,k,y)}$) has been adjusted to 90% in accordance with Clarification 2 of VM0050 v1.0. The Project Proponent has provided justification demonstrating compliance with all three required customer support criteria, including the selection of the Tramontana ICS based on local cooking needs, provision of user materials in the local language, and establishment of local maintenance teams and accessible support channels. - QA/QC procedures have also been adequately addressed. These include field staff training prior to monitoring, direct access for end users to experienced local staff for issue resolution, use of oversampling to exceed confidence thresholds, and internal checks covering the entire population annually, including revisits to sampled households to verify data accuracy. The above clarifications are deemed reasonable and acceptable; hence, this CL is closed.	

CL ID	05	Section no.	5.3.2 PD	Date: 10/07/2025
Description of CL				
In section 5.3.2 of the PD, - It states that the parameter $\eta_{new,avg,y}$ in Table 11 is monitored using the Water Boiling Test Protocol v4.0. The PP is to justify using this older version of the protocol, while the Clean Cooking Alliance (2014) Water Boiling Test Protocol v4.2.3 is the latest publicly available version. - It is stated that 1,793 baseline samples were used to calculate the required sample size based on a 90/10 confidence/precision threshold. However, another section of the document references 1,944 baseline samples collected for the first project instance. The PP is requested to clarify on this discrepancy and confirm the accurate number of baseline samples used in the baseline analysis.				
Project proponent response				Date: 01/08/2025
- Water Boiling Test Protocol v4.2.3 has been used. This has been corrected in section 5.3.2 of the PD and can also be confirmed with the test report presented in Appendix 2 1,944 is the actual number of baseline surveys completed for the first project instance. The number 1,793 however is referring to preliminary fuel consumption data which was used to calculate the minimum sample size requirement for the KPT and baseline survey. Within PD section 5.3.2.2, the approach and calculations used to work out the minimum sample size of parameters $BC_{b,i,y}$, $BC_{p,j,k,y}$ and $Hh_i / Hh_{j,k}$ have been rewritten to improve clarity.				
Documentation provided by project proponent				
VCS4457 Project Description v1.7				
VVB assessment				Date: 07/08/2025

1. The VVB has reviewed the revised PD v1.7 (Section 5.3.2) and confirms that the reference to the Water Boiling Test Protocol has been corrected to version 4.2.3 (Clean Cooking Alliance, 2014). Use of the correct version has been verified through the efficiency test report provided in Appendix 2.
2. The VVB has also reviewed the clarification regarding the baseline sample size in revised PD v1.7 (Section 5.3.2). The Project Proponent has confirmed that 1,944 baseline surveys were conducted, while the figure 1,793 refers to preliminary fuel consumption data used for calculating the minimum required sample size. Section 5.3.2.2 has been updated to improve transparency on the sampling methodology.

The above clarifications are deemed reasonable and acceptable; hence, this CL is closed.

CL ID	06	Section no.	PD 5.2	Date: 10/07/2025
Description of CL				
The measurement method, to determine the value for parameter $\eta_{new,avg,y}$, is as per CDM Standard for Sampling and Surveys for CDM Project Activities and Programmes of Activities v9.0 (option 1 of the applied methodology VM0050, v1.0); however it is not clear how the required sample size has been attained and whether this complies with para 14 of the aforementioned sampling standard which states, "If the parameter of interest is a numeric mean value (i.e. not a proportion or percentage) the Student's t-distribution shall be used if the resulting sample size is less than 30". PP is requested to clarify on this and provide reliability test (to demonstrate 90/10 confidence and precision levels are met) and sample size calculation sheet.				
Project proponent response				Date: 01/08/2025
To determine the value of $\eta_{new,avg,y}$ at $y=0$, option 2 of the VM0050 v1.0 methodology has been used: "Manufacturer-certified value that is determined via Water Boiling Test". The manufacturer brochure stating the efficiency has been provided to the VVB and the WBT test results from which the efficiency has been obtained is presented in Appendix 2 of the PD. As option 2 of the methodology has been used, analysis of the parameter in line with the CDM Standards is not applicable.				
Documentation provided by project proponent				
CRS-23 Manufacturer Brochure				
VVB assessment				Date: 07/08/2025
The VVB has reviewed the revised PD v1.7 (Section 5.2 and Section 5.3.2.1) and confirms that the Project Proponent has included a clarification stating that Option 2 of the applied methodology VM0050 v1.0 was used to determine the parameter $\eta_{new,avg,y}$ at $y=0$, based on a manufacturer-certified value obtained via Water Boiling Test (WBT). The reported efficiency is supported by the manufacturer's brochure and corresponding WBT results, provided in Appendix 2 of the PD. As per the methodology, when Option 2 is applied, compliance with the CDM Standard for Sampling and Surveys is not required. The above clarification is deemed reasonable and acceptable; hence, this CL is closed.				

Table 3. CARs from this verification

CAR ID	01	Section no.	1.2 PD	Date: 10/07/2025
Description of CAR				
Section 1.2 of the PD does not reflect the methodology change and requantification for the first monitoring period. PP shall revise the PD accordingly.				
Project proponent response				Date: 01/08/2025
Section 1.2 of the PD has been updated to include the Requantification and Methodology change for the first monitoring period.				
Documentation provided by project proponent				
VCS4457 Project Description v1.7				
VVB assessment				Date: 07/08/2025
The VVB has reviewed the revised PD and confirms that section 1.2 has been updated to reflect the methodology change and requantification for the first monitoring period as required. Hence this CAR is closed.				

CAR ID	02	Section no.	1.11 PD	Date: 10/07/2025
Description of CAR				
In section 1.11 of the PD, the date format used in this section is not in line with the VCS Project Description Template v4.4, which specifies the use of the 'DDMMYYYY' format. The PP shall revise the date formatting accordingly and ensure that the correct format is applied consistently across all relevant sections.				
Project proponent response				Date: 01/08/2025
Section 1.11 of the PD has been updated to reflect the DD-Month-YYYY format in line with the VCS Project Description Template v4.4.				
The two tables in Section 4.4 have also been updated to the DD-MMM-YYYY format in line with the VCS Project Description Template v4.4.				
Documentation provided by project proponent				
VCS4457 Project Description v1.7				
VVB assessment				Date: 07/08/2025
The VVB has reviewed the revised PD and confirms that all relevant sections have been updated to apply the correct date format (DD-MMM-YYYY) in accordance with the VCS Project Description Template v4.4 instructions. Hence, this CAR is closed.				

CAR ID	03	Section no.	5.1 PD	Date: 10/07/2025
Description of CAR				
In section 5.1 of the PD, for all parameters listed: 1. The source field references the 'most recent version of IPCC Guidelines' or 'Option 1 as specified in the methodology' without specifying the exact version. 2. The justification field is left blank. As required by the VCS Project Description Template and VM0050, justification must be provided. The PP shall address the above and revise the PD and the relevant section(s) of the Requantification Report in a similar manner.				
Project proponent response				Date: 01/08/2025
1. Across all parameters in section 5.1 where the IPCC Guidelines have been referenced, the “source of data” section has been updated to include the specific version and year of IPCC Guidelines used. A reference and link to the version has also been included as a footnote. 2. The justification fields for the parameters in section 5.1 have been updated in line with the VCS Project Description Template v4.4 to include relevant details on data description and measurement methods. Note that the equivalent section in the Requantification Report has also been updated to include the changes to the parameters described above.				
Documentation provided by project proponent				
• VCS4457 Project Description v1.7 • VCS4457 Requantification Report v1.1				
VVB assessment				Date: 07/08/2025
The VVB has reviewed the revised PD and revised Requantification Report and confirms that section 5.1 of the PD and section 3.1 of the Requantification Report have been updated to specify the exact version and year of the IPCC Guidelines used, with proper references included as footnotes. Justification fields for all parameters have also been completed in accordance with the VCS Project Description Template v4.4, detailing data sources and measurement approaches. Hence, this CAR is closed.				

CAR ID	04	Section no.	5.2 PD	Date: 10/07/2025
Description of CAR				
In section 5.2 of the PD, 1. For parameter $f_{NRB,y}$, the monitoring equipment section is blank. Even though the parameter is desktop-derived, please provide a description of datasets, such as GIS sources, remote sensing imagery, or biomass inventory studies used in the estimation process. Specify any models or third-party tools involved. Furthermore, QA/QC procedures are missing. Please outline the protocols in place to verify the quality of NRB and RB inputs, such as expert reviews, data triangulation, or comparison with historical datasets. 2. For parameter $\eta_{new,avg,y}$, the report does not provide details on the calibration of laboratory equipment used in the Water Boiling Tests. The Project Proponent (PP) shall confirm whether the laboratories follow ISO or equivalent national calibration standards and whether the equipment is periodically certified 3. For parameter $BC_{p,j,k,y}$, the QA/QC section does not specify whether or how the digital scales and moisture meters used in fieldwork are calibrated. The Project Proponent (PP) shall describe the calibration protocol				

Project proponent response	Date: 01/08/2025
1. The monitoring equipment section has been updated to include details on the methods and data used in the calculation of fNRB. Specifically, the KPT process used to measure wood consumption has been described and sources including IPCC Guidelines and state reports have been listed. Note that the complete list of data sources and specific methods used, including references and detailed calculations for the first project instance, have been described in the spreadsheet "ICS-01 fNRB Calculation Instance 1". 2. The WBTs are conducted by experienced laboratories following ISO standards. Details of this have been added into the QA/QC section of the $\eta_{new,avg,y}$ parameter table and ISO certificates from the manufacturer and WBT testing company have been provided to the VVB. 3. The digital scales and moisture meters have been calibrated by the respective device manufacturers. This protocol has been added into the QA/QC section of the relevant parameter tables.	
Documentation provided by project proponent	
• Moisture Meter Manual • Digital Scales Folder • VCS4457 Project Description v1.7	
VVB assessment	Date: 07/08/2025
1. The VVB has reviewed the revised PD (section 5.2) and confirms that the Project Proponent has updated the monitoring equipment and QA/QC procedures for the parameter $f_{NRB,y}$. The PD now clearly outlines the data sources and measurement approaches, including the use of KPT tests, IPCC Guidelines, national forest and health surveys, and census data. References and detailed calculations are transparently provided in the spreadsheet "ICS-01 fNRB Calculation Instance 1." QA/QC measures are also adequately described, including crosschecks with government publications and academic research. 2. The VVB has reviewed the revised PD (sections 5.2 and 5.3.2.1) and confirms that the QA/QC procedures for the parameter $\eta_{new,avg,y}$ have been adequately updated. The revised PD confirms that Water Boiling Tests (WBTs) are conducted by experienced laboratories adhering to ISO standards. The VVB has also reviewed the ISO certificates submitted for both the manufacturer and the testing laboratory, confirming that the laboratory equipment is calibrated in accordance with international standards and that the testing procedures meet the applicable quality requirements. 3. The VVB has reviewed the revised PD and confirms that the QA/QC procedures for the parameter $BC_{p,j,k,y}$ have been updated to include confirmation that the digital scales and moisture meters used during fieldwork have been calibrated by the respective manufacturers. This information has been appropriately reflected in the relevant parameter table under Section 5.2. The above clarifications are deemed reasonable and acceptable; hence this CAR is closed.	

CAR ID	05	Section no.	5.3.2 PD	Date: 10/07/2025
Description of CAR				
In section 5.3.2 of the PD, 1. The report states that the Water Boiling Test Protocol v4.0 was used to monitor the parameter $\eta(new, avg, y)$. However, based on the evidence provided by the Project Proponent (PP), the test was conducted using the Clean Cooking Alliance's Water Boiling Test Protocol v4.2.3 (2014), which is the latest available version. PP shall correct the report to accurately reflect the version used. 2. Under 'Implementation plan & data quality control', it outlines general field procedures and staff training but does not specify who is responsible for the final verification and approval of field data before inclusion in monitoring reports. It also lacks reference to version control or audit mechanisms. The Project Proponent (PP) shall clarify the following: • Who is responsible for final data validation and approval (e.g., specific personnel, roles, or oversight structure)? • What version control mechanisms exist within the Tramontana ICS system or other data tools to ensure traceability of original versus updated data? • How are errors or updates identified, logged, and reported to maintain transparency and accountability during third-party verification?				
Project proponent response				Date: 01/08/2025
1. Water Boiling Test Protocol v4.2.3 has been used. This has been corrected in section 5.3.2 of the PD and can also be confirmed with the test report presented in Appendix 2. 2. Section 5.3.2.3 has been updated to include further details on our data validation and approval process. This includes what the particular roles of the field, Hyderabad and London offices are and specifically how data is processed internally before final output and approval. Details of the version control mechanisms have also been added into the section and includes a description of the mechanisms employed to ensure traceability and how the office teams utilise them on a daily basis. Finally, details have been added to explain the methods used for identifying, logging and reporting any updates or errors.				
Documentation provided by project proponent				
VCS4457 Project Description v1.7				
VVB assessment				Date: 07/08/2025
1. The VVB has reviewed the revised PD v1.7 and confirms that Section 5.3.2 has been corrected to reflect the use of the Clean Cooking Alliance's Water Boiling Test Protocol v4.2.3 (2014), in alignment with the supporting test reports provided in Appendix 2. 2. Additionally, Section 5.3.2.3 has been updated to include detailed descriptions of the data validation and approval structure, specifying the responsibilities of field teams and Tramontana offices in Hyderabad and London. The PD also outlines version control mechanisms implemented within the Tramontana ICS system, and describes procedures for identifying, logging, and reporting errors or data updates to ensure traceability and transparency during monitoring and verification. The above clarifications are deemed reasonable and acceptable; hence this CAR is closed.				

CAR ID	06	Section no.	1.1 RR	Date: 10/07/2025
Description of CAR				
In section 1,2 of the Requantification Report, 1. The Audit History section does not comply with Section 1.1 of the VCS Requantification Report Template, as it omits prior validation and verification audits. PP shall revise the section to include all relevant historical audits, including the Requantification Approval Request. 2. The VVB name associated with the requantification audit is incorrectly stated. 3. The inclusion of the 2nd Monitoring Period under the Audit section is inaccurate, as this period has not undergone verification and pertains to a future monitoring period. PP shall remove or appropriately revise this entry. As per the VCS Methodology Change and Requantification Procedure v4.0, the methodology change and requantification request is a stand-alone submission and should not be clubbed with future verification review. PP shall correct the report accordingly.				
Project proponent response				Date: 01/08/2025
1. Section 1.2 has been updated to include the historical audit and Requantification audit 2. For the Requantification audit, the VVB name has been updated 3. The audit for the second monitoring period has been removed				
Documentation provided by project proponent				
VCS4457 Requantification Report v1.1				
VVB assessment				Date: 07/08/2025
The VVB has reviewed the revised Requantification Report and confirms the following corrections have been made in accordance with the VCS Methodology Change and Requantification Procedure v4.0: 1. Audit History: Section 1.2 has been appropriately updated to include the complete audit history, including the original validation, prior verification(s), and the current Requantification Approval Request, as required by the VCS Requantification Report Template v4.0. 2. Correction of VVB Name: The name of the Validation/Verification Body associated with the current requantification audit has been corrected to reflect the accurate entity. 3. Removal of Future Monitoring Period: The previously referenced second monitoring period has been removed from the audit history, in line with the requirement that the Requantification Request remains distinct from future monitoring and verification activities. The corrections provided are complete and align with applicable VCS guidance. Hence, this CAR is closed.				

CAR ID	07	Section no.	3.3 RR	Date: 10/07/2025
Description of CAR				
In section 3.3 of the Requantification Report, 1. In accordance with Section 2.3.5 of the VCS Methodology Change and Requantification Procedure v4.0, the project proponent is required to revise and update key sections of the project documentation, including Section 3.3 – Monitoring Plan. 2. As per Section 3.3 of the Requantification Report Template, the Monitoring Plan must clearly describe the process and schedule followed during the monitoring period for obtaining, compiling, and analyzing monitored data and parameters detailed in Section 4.2. The current submission does not meet this requirement and must be revised accordingly. 3. The current Section 3.3 lacks sufficient detail regarding data collection methodology, roles and responsibilities of personnel, internal audit procedures, sampling implementation, and QA/QC protocols. PP shall revise and complete this section to ensure clarity, completeness, and transparency in the monitoring approach adopted for the requantification period. PP is requested to comply with the above.				
Project proponent response				Date: 01/08/2025
1. Section 3.3 has been updated to include details on the Monitoring Plan. This section and the rest of the Requantification Report now complies with section 2.3.5 of the VCS Methodology Change and Requantification Procedure v4.0. 2. Section 3.3.1 has been updated to include extensive information on the data recording and compiling process followed during the monitoring period. The section includes details on the data collected, how it is stored and the proprietary technology used to ensure data accuracy and security. Section 3.3.2.1 has also been updated with details on how each of the monitored parameters are sampled and obtained in the field. 3. Further to the above, Section 3.3.2.1 has been updated to include details on the sampling design and QA/QC procedures for each of the parameters monitored during the period. The section also now includes statistical analysis to demonstrate how methodology and CDM sampling requirements are met. Section 3.3.2.2 presents the sample size calculations for each of the monitored parameters and explains any assumptions or additional data sources utilised. Details on project implementation and data quality control including a description of the roles and responsibilities of the different office teams (field, Hyderabad, London) has been described in section 3.3.2.3. The section also discusses the frequency of monitoring as well as provides a detailed explanation of the implementation strategy employed for the project.				
Documentation provided by project proponent				
VCS4457 Requantification Report v1.1				
VVB assessment				Date: 07/08/2025

The VVB has reviewed the revised Requantification Report and confirms that Section 3.3 has been updated in line with the requirements of the VCS Methodology Change and Requantification Procedure v4.0 and the Requantification Report Template.

1. Section 3.3 of the Requantification Report has been updated to clearly describe the monitoring approach applied during the requantification period, in line with the requirements of Section 2.3.5 of the VCS Methodology Change and Requantification Procedure v4.0.
2. Section 3.3.1 now provides a comprehensive overview of the monitoring process, including detailed information on how data was obtained, recorded, compiled, and analyzed. The section also describes the proprietary systems used for data storage and management, with appropriate safeguards to ensure data integrity.
3. Section 3.3.2.1 includes a detailed explanation of the data collection methodology, including field protocols for each parameter, sample design, and implementation procedures. The section adequately addresses QA/QC protocols, including statistical analyses used to confirm sampling requirements. Additionally, Section 3.3.2.2 presents the sample size calculations and underlying assumptions used, while Section 3.3.2.3 clearly outlines the roles and responsibilities of field teams and oversight offices in Hyderabad and London. The internal data validation process, frequency of monitoring, and implementation strategy are also described in sufficient detail.

The revised monitoring plan demonstrates adequate clarity, completeness, and transparency, and meets the expectations set forth under the applicable VCS procedural documents. Hence, this CAR is closed.

CAR ID	08	Section no.	3.3 RR	Date: 10/07/2025
Description of CAR				
In section 4.4 of the Requantification Report, in the table comparing ex-ante and achieved emissions reductions, the reported ex-ante value for EC _y is not consistent with the PD. The PP is requested to clarify on this discrepancy and rectify as required.				
Project proponent response				Date: 01/08/2025
The value of EC _y in section 4.4 has been corrected and is now consistent with the PD.				
Documentation provided by project proponent				
VCS4457 Requantification Report v1.1				
VVB assessment				Date: 07/08/2025
The VVB has reviewed the updated Requantification Report and confirms that the value of EC _y in section 4.4 has been corrected to align with the Project Description. The discrepancy between the ex-ante emissions reductions reported in the Requantification Report and the PD has been adequately addressed. Hence, this CAR is closed.				

Table 4. FARs from this verification

FAR ID	01	Section no.		Date: 07/08/2025
Description of FAR				
The estimated emission reductions are to be made consistent between the Verra project page and the project documents. VVB to verify the same in the subsequent verification.				
Project proponent response				Date: DD/MM/YYYY

Documentation provided by project proponent	
VVB assessment	Date: DD/MM/YYYY

FAR ID	02	Section no.		Date: 07/08/2025
Description of FAR				
PP to comply with para 6.2 of VMR0006 v1.2 in conjunction with correction 1 of Corrections & Clarifications to VMR0006 v1.0 and must conduct a simple random sampling where a survey covers two or more project activity instances and the population is homogeneous, and where the population is not homogenous, stratified random sampling must be used. Also, follow-up baseline surveys must be conducted at most every five years from the date of the last survey in control households that do not participate in the project. Moreover, baseline survey performed prior to crediting period renewal must be conducted as physical on-site visit. VVV to verify this during subsequent verifications. FAR 02 from this verification is raised.				
Project proponent response				Date: DD/MM/YYYY
Documentation provided by project proponent				
VVB assessment				Date: DD/MM/YYYY